

As we make the time interval smaller and smaller we find that the average velocity is getting closer and closer to 9.8. We can be a little more precise by finding the average velocity between $t = 1$ and $t = 1 + h$ — this is very similar to what we did for tangent lines.

$$\begin{aligned}\text{average velocity} &= \frac{s(1+h) - s(1)}{(1+h) - 1} \\ &= \frac{4.9(1+h)^2 - 4.9}{h} \\ &= \frac{9.8h + 4.9h^2}{h} \\ &= 9.8 + 4.9h\end{aligned}$$

Now as we squeeze this window between $t = 1$ and $t = 1 + h$ down towards zero, the average velocity becomes the “instantaneous velocity” — just as the slope of the secant line becomes the slope of the tangent line. This is our second limit

$$v(1) = \lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = 9.8$$

More generally we define the instantaneous velocity at time $t = a$ to be the limit

$$v(a) = \lim_{h \rightarrow 0} \frac{s(a+h) - s(a)}{h}$$

We read this as

The velocity at time a is equal to the limit as h goes to zero of $\frac{s(a+h) - s(a)}{h}$.

While we have solved the problem stated at the start of this section, it is clear that if we wish to solve similar problems that we will need to understand limits in a more general and systematic way.

1.3 ▲ The Limit of a Function

Before we come to definitions, let us start with a little notation for limits.

Notation 1.3.1.

We will often write

$$\lim_{x \rightarrow a} f(x) = L$$

which should be read as

The limit of $f(x)$ as x approaches a is L .

The notation is just shorthand — we don't want to have to write out long sentences as we do our mathematics. Whenever you see these symbols you should think of that sentence.

This shorthand also has the benefit of being mathematically precise (we'll see this later), and (almost) independent of the language in which the author is writing. A mathematician who does not speak English can read the above formula and understand exactly what it means.

In mathematics, like most languages, there is usually more than one way of writing things and we can also write the above limit as

$$f(x) \rightarrow L \text{ as } x \rightarrow a$$

This can also be read as above, but also as

$$f(x) \text{ goes to } L \text{ as } x \text{ goes to } a$$

They mean exactly the same thing in mathematics, even though they might be written, read and said a little differently.

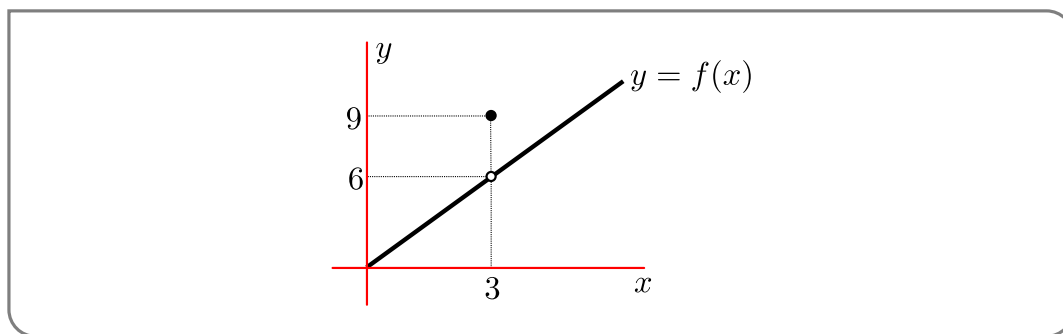
To arrive at the definition of limit, we want to start⁶ with a very simple example.

Example 1.3.2

Consider the following function.

$$f(x) = \begin{cases} 2x & x < 3 \\ 9 & x = 3 \\ 2x & x > 3 \end{cases}$$

This is an example of a piece-wise function⁷. That is, a function defined in several pieces, rather than as a single formula. We evaluate the function at a particular value of x on a case-by-case basis. Here is a sketch of it



Notice the two circles in the plot. One is open, $\circ$ and the other is closed $\bullet$.

- A filled circle has quite a precise meaning — a filled circle at (x, y) means that the function takes the value $f(x) = y$.

6 Well, we had two limits in the previous sections, so perhaps we really want to “restart” with a very simple example.

7 We saw another piecewise function back in Example [0.4.5](#).

- An open circle is a little harder — an open circle at $(3, 6)$ means that the point $(3, 6)$ is not on the graph of $y = f(x)$, i.e. $f(3) \neq 6$. We should only use the open circle where it is absolutely necessary in order to avoid confusion.

This function is quite contrived, but it is a very good example to start working with limits more systematically. Consider what the function does close to $x = 3$. We already know what happens exactly at 3 — $f(3) = 9$ — but I want to look at how the function behaves very close to $x = 3$. That is, what does the function do as we look at a point x that gets closer and closer to $x = 3$.

If we plug in some numbers very close to 3 (but not exactly 3) into the function we see the following:

x	2.9	2.99	2.999	$\circ$	3.001	3.01	3.1
$f(x)$	5.8	5.98	5.998	$\circ$	6.002	6.02	6.2

So as x moves closer and closer to 3, without being exactly 3, we see that the function moves closer and closer to 6. We can write this as

$$\lim_{x \rightarrow 3} f(x) = 6$$

That is

The limit as x approaches 3 of $f(x)$ is 6.

So for x very close to 3, without being exactly 3, the function is very close to 6 — which is a long way from the value of the function exactly at 3, $f(3) = 9$. Note well that the behaviour of the function as x gets very close to 3 *does not* depend on the value of the function *at* 3.

Example 1.3.2

We now have enough to make an informal definition of a limit, which is actually sufficient for most of what we will do in this text.

Definition 1.3.3 (Informal definition of limit).

We write

$$\lim_{x \rightarrow a} f(x) = L$$

if the value of the function $f(x)$ is sure to be arbitrarily close to L whenever the value of x is close enough to a , without⁸ being exactly a .

8 You may find the condition “without being exactly a ” a little strange, but there is a good reason for it. One very important application of limits, indeed the main reason we teach the topic, is in the definition of derivatives (see Definition 2.2.1 in the next chapter). In that definition we need to compute the limit $\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$. In this case the function whose limit is being taken, namely $\frac{f(x) - f(a)}{x - a}$, is not defined at all at $x = a$.

In order to make this definition more mathematically correct, we need to make the idea of “closer and closer” more precise — we do this in Section 1.7. It should be emphasised that the formal definition and the contents of that section are optional material.

For now, let us use the above definition to examine a more substantial example.

Example 1.3.4

Let $f(x) = \frac{x-2}{x^2+x-6}$ and consider its limit as $x \rightarrow 2$.

- We are really being asked

$$\lim_{x \rightarrow 2} \frac{x-2}{x^2+x-6} = \text{what?}$$

- Now if we try to compute $f(2)$ we get $0/0$ which is undefined. The function is not defined at that point — this is a good example of why we need limits. We have to sneak up on these places where a function is not defined (or is badly behaved).
- **VERY IMPORTANT POINT:** the fraction $\frac{0}{0}$ is *not* ∞ and it is not 1, it is not defined. We cannot ever divide by zero in normal arithmetic and obtain a consistent and mathematically sensible answer. If you learned otherwise in high-school, you should quickly unlearn it.
- Again, we can plug in some numbers close to 2 and see what we find

x	1.9	1.99	1.999	$\circ$	2.001	2.01	2.1
$f(x)$	0.20408	0.20040	0.20004	$\circ$	0.19996	0.19960	0.19608

- So it is reasonable to suppose that

$$\lim_{x \rightarrow 2} \frac{x-2}{x^2+x-6} = 0.2$$

Example 1.3.4

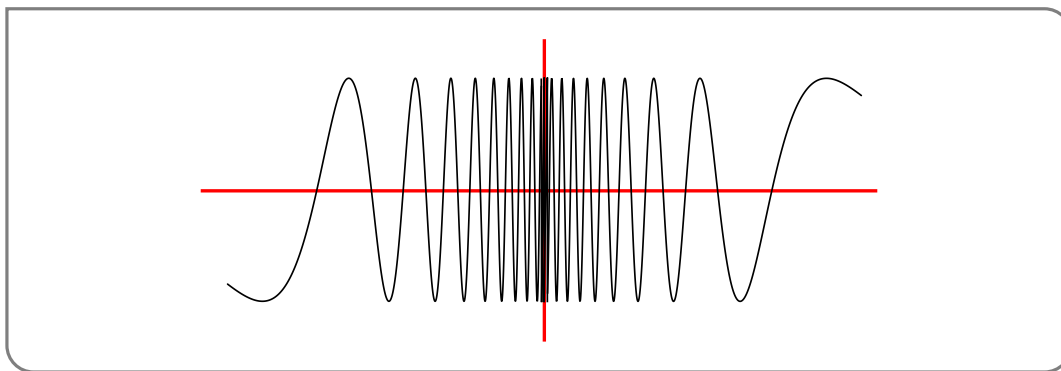
The previous two examples are nicely behaved in that the limits we tried to compute actually exist. We now turn to two nastier examples⁹ in which the limits we are interested in do not exist.

Example 1.3.5 (A bad example)

Consider the following function $f(x) = \sin(\pi/x)$. Find the limit as $x \rightarrow 0$ of $f(x)$.

We should see something interesting happening close to $x = 0$ because $f(x)$ is undefined there. Using your favourite graph-plotting software you can see that the graph looks roughly like

⁹ Actually, they are good examples, but the functions in them are nastier.



How to explain this? As x gets closer and closer to zero, π/x becomes larger and larger (remember what the plot of $y = 1/x$ looks like). So when you take sine of that number, it oscillates faster and faster the closer you get to zero. Since the function does not approach a single number as we bring x closer and closer to zero, the limit does not exist.

We write this as

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right) \text{ does not exist}$$

It's not very inventive notation, however it is clear. We frequently abbreviate "does not exist" to "DNE" and rewrite the above as

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right) = \text{DNE}$$

Example 1.3.5

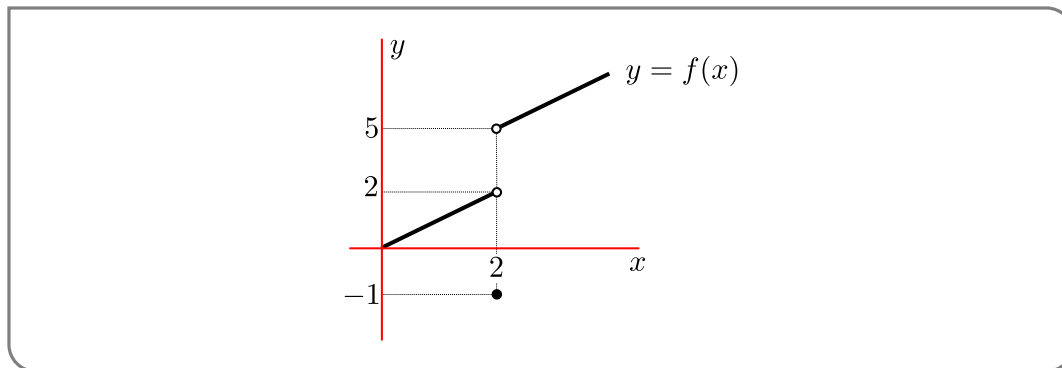
In the following example, the limit we are interested in does not exist. However the way in which things go wrong is quite different from what we just saw.

Example 1.3.6

Consider the function

$$f(x) = \begin{cases} x & x < 2 \\ -1 & x = 2 \\ x + 3 & x > 2 \end{cases}$$

- The plot of this function looks like this



- So let us plug in numbers close to 2.

x	1.9	1.99	1.999	$\circ$	2.001	2.01	2.1
$f(x)$	1.9	1.99	1.999	$\circ$	5.001	5.01	5.1

- This isn't like before. Now when we approach from below, we seem to be getting closer to 2, but when we approach from above we seem to be getting closer to 5. Since we are not approaching the same number the limit does not exist.

$$\lim_{x \rightarrow 2} f(x) = \text{DNE}$$

Example 1.3.6

While the limit in the previous example does not exist, the example serves to introduce the idea of “one-sided limits”. For example, we can say that

As x moves closer and closer to two *from below* the function approaches 2.

and similarly

As x moves closer and closer to two *from above* the function approaches 5.

Definition 1.3.7 (Informal definition of one-sided limits).

We write

$$\lim_{x \rightarrow a^-} f(x) = K$$

when the value of $f(x)$ gets closer and closer to K when $x < a$ and x moves closer and closer to a . Since the x -values are always less than a , we say that x approaches a *from below*. This is also often called the left-hand limit since the x -values lie to the left of a on a sketch of the graph.

We similarly write

$$\lim_{x \rightarrow a^+} f(x) = L$$

when the value of $f(x)$ gets closer and closer to L when $x > a$ and x moves closer and closer to a . For similar reasons we say that x approaches a *from above*, and sometimes refer to this as the right-hand limit.

Note — be careful to include the superscript $+$ and $-$ when writing these limits. You might also see the following notations:

$$\begin{aligned} \lim_{x \rightarrow a^+} f(x) &= \lim_{x \rightarrow a+} f(x) = \lim_{x \downarrow a} f(x) = \lim_{x \searrow a} f(x) = L && \text{right-hand limit} \\ \lim_{x \rightarrow a^-} f(x) &= \lim_{x \rightarrow a-} f(x) = \lim_{x \uparrow a} f(x) = \lim_{x \nearrow a} f(x) = L && \text{left-hand limit} \end{aligned}$$

but please use with the notation in Definition 1.3.7 above.

Given these two similar notions of limits, when are they the same? The following theorem tell us

Theorem 1.3.8 (Limits and one sided limits).

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Notice that this is really two separate statements because of the “if and only if”

- If the limit of $f(x)$ as x approaches a exists and is equal to L , then both the left-hand and right-hand limits exist and are equal to L . AND,
- If the left-hand and right-hand limits as x approaches a exist and are equal, then the limit as x approaches a exists and is equal to the one-sided limits.

That is — the limit of $f(x)$ as x approaches a will only exist if it doesn't matter which way we approach a (either from left or right) AND if we get the same one-sided limits when we approach from left and right, then the limit exists.

We can rephrase the above by writing the contrapositives¹⁰ of the above statements.

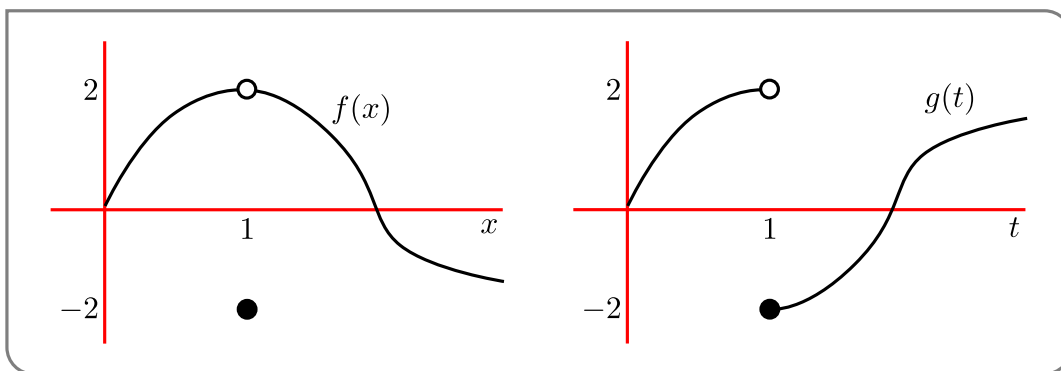
- If either of the left-hand and right-hand limits as x approaches a fail to exist, or if they both exist but are different, then the limit as x approaches a does not exist. AND,
- If the limit as x approaches a does not exist, then the left-hand and right-hand limits are either different or at least one of them does not exist.

Here is another limit example

Example 1.3.9

Consider the following two functions and compute their limits and one-sided limits as x approaches 1:

10 Given a statement of the form “If A then B”, the contrapositive is “If not B then not A”. They are logically equivalent — if one is true then so is the other. We must take care not to confuse the contrapositive with the converse. Given “If A then B”, the converse is “If B then A”. These are definitely not the same. To see this consider the statement “If he is Shakespeare then he is dead.” The converse is “If he is dead then he is Shakespeare” — clearly garbage since there are plenty of dead people who are not Shakespeare. The contrapositive is “If he is not dead then he is not Shakespeare” — which makes much more sense.



These are a little different from our previous examples, in that we do not have formulas, only the sketch. But we can still compute the limits.

- Function on the left — $f(x)$:

$$\lim_{x \rightarrow 1^-} f(x) = 2$$

$$\lim_{x \rightarrow 1^+} f(x) = 2$$

so by the previous theorem

$$\lim_{x \rightarrow 1} f(x) = 2$$

- Function on the right — $g(t)$:

$$\lim_{t \rightarrow 1^-} g(t) = 2$$

$$\text{and } \lim_{t \rightarrow 1^+} g(t) = -2$$

so by the previous theorem

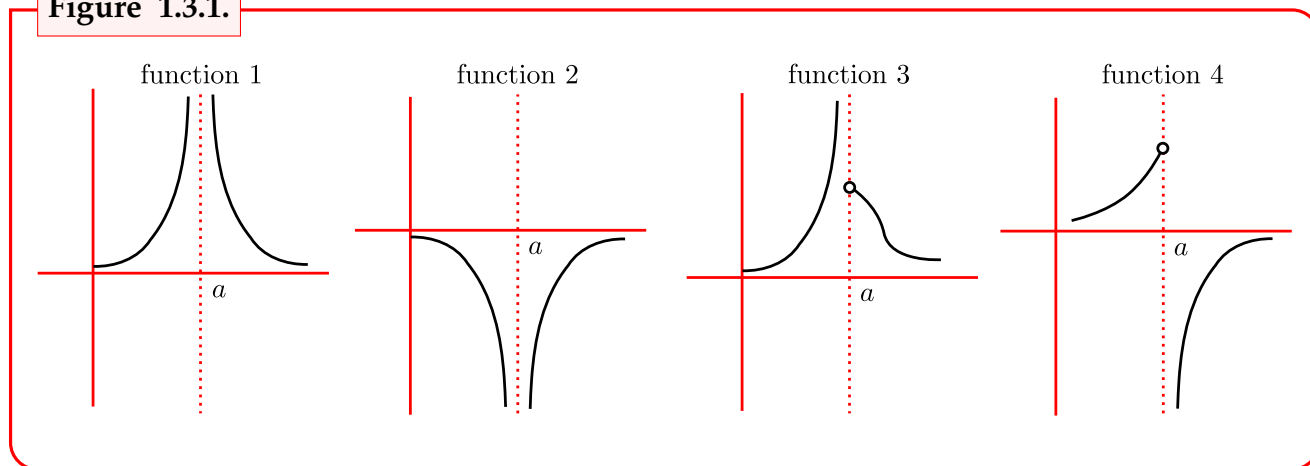
$$\lim_{t \rightarrow 1} g(t) = \text{DNE}$$

Example 1.3.9

We have seen 2 ways in which a limit does not exist — in one case the function oscillated wildly, and in the other there was some sort of “jump” in the function, so that the left-hand and right-hand limits were different.

There is a third way that we must also consider. To describe this, consider the following four functions:

Figure 1.3.1.



None of these functions are defined at $x = a$, nor do the limits as x approaches a exist. However we can say more than just “the limits do not exist”.

Notice that the value of function 1 can be made bigger and bigger as we bring x closer and closer to a . Similarly the value of the second function can be made arbitrarily large and negative (i.e. make it as big a negative number as we want) by bringing x closer and closer to a . Based on this observation we have the following definition.

Definition 1.3.10.

We write

$$\lim_{x \rightarrow a} f(x) = +\infty$$

when the value of the function $f(x)$ becomes arbitrarily large and positive as x gets closer and closer to a , without being exactly a .

Similarly, we write

$$\lim_{x \rightarrow a} f(x) = -\infty$$

when the value of the function $f(x)$ becomes arbitrarily large and negative as x gets closer and closer to a , without being exactly a .

A good examples of the above is

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$$

$$\lim_{x \rightarrow 0} -\frac{1}{x^2} = -\infty$$

IMPORTANT POINT: Please do not think of “ $+\infty$ ” and “ $-\infty$ ” in these statements as numbers. You should think of $\lim_{x \rightarrow a} f(x) = +\infty$ and $\lim_{x \rightarrow a} f(x) = -\infty$ as special cases of $\lim_{x \rightarrow a} f(x) = \text{DNE}$. The statement

$$\lim_{x \rightarrow a} f(x) = +\infty$$

does not say “the limit of $f(x)$ as x approaches a is positive infinity”. It says “the function $f(x)$ becomes arbitrarily large as x approaches a ”. These are different statements; remember that ∞ is not a number¹¹

Now consider functions 3 and 4 in Figure 1.3.1. Here we can make the value of the function as big and positive as we want (for function 3) or as big and negative as we want (for function 4) but only when x approaches a from one side. With this in mind we can construct similar notation and a similar definition:

11 One needs to be very careful making statements about infinity. At some point in our lives we get around to asking ourselves “what is the biggest number”, and we realise there isn’t one. That is, we can go on counting integer after integer, for ever and not stop. Indeed the set of integers is the first infinite thing we really encounter. It is an example of a *countably infinite* set. The set of real-numbers is actually much bigger and is *uncountably infinite*. In fact there are an infinite number of different sorts of infinity! Much of the theory of infinite sets was developed by Georg Cantor; we mentioned him back in Section 0.2 and he is well worth googling.

Definition 1.3.11.

We write

$$\lim_{x \rightarrow a^+} f(x) = +\infty$$

when the value of the function $f(x)$ becomes arbitrarily large and positive as x gets closer and closer to a from above (equivalently — from the right), without being exactly a .

Similarly, we write

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

when the value of the function $f(x)$ becomes arbitrarily large and negative as x gets closer and closer to a from above (equivalently — from the right), without being exactly a .

The notation

$$\lim_{x \rightarrow a^-} f(x) = +\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

has a similar meaning except that limits are approached from below / from the left.

So for function 3 we have

$$\lim_{x \rightarrow a^-} f(x) = +\infty$$

$$\lim_{x \rightarrow a^+} f(x) = \text{some positive number}$$

and for function 4

$$\lim_{x \rightarrow a^-} f(x) = \text{some positive number}$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

More examples:

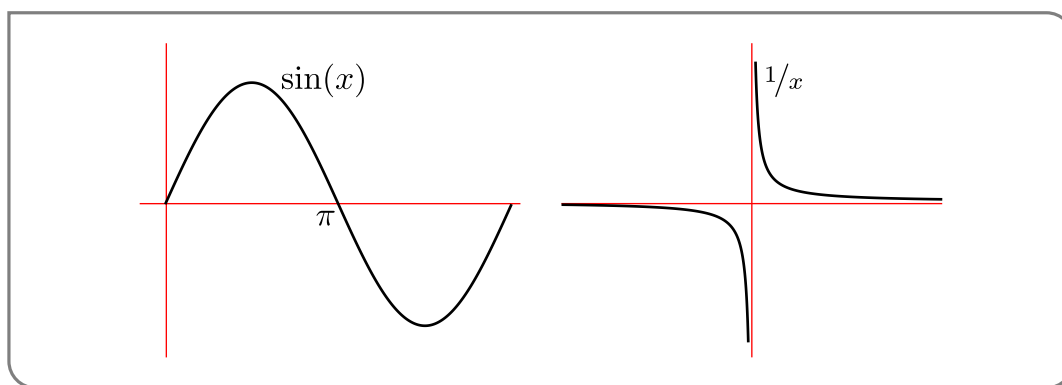
Example 1.3.12

Consider the function

$$g(x) = \frac{1}{\sin(x)}$$

Find the one-sided limits of this function as $x \rightarrow \pi$.

Probably the easiest way to do this is to first plot the graph of $\sin(x)$ and $1/x$ and then think carefully about the one-sided limits:



- As $x \rightarrow \pi$ from the left, $\sin(x)$ is a small positive number that is getting closer and closer to zero. That is, as $x \rightarrow \pi^-$, we have that $\sin(x) \rightarrow 0$ through positive numbers (i.e. from above). Now look at the graph of $1/x$, and think what happens as we move $x \rightarrow 0^+$, the function is positive and becomes larger and larger.

So as $x \rightarrow \pi$ from the left, $\sin(x) \rightarrow 0$ from above, and so $1/\sin(x) \rightarrow +\infty$.

- By very similar reasoning, as $x \rightarrow \pi$ from the right, $\sin(x)$ is a small negative number that gets closer and closer to zero. So as $x \rightarrow \pi$ from the right, $\sin(x) \rightarrow 0$ through negative numbers (i.e. from below) and so $1/\sin(x) \rightarrow -\infty$.

Thus

$$\lim_{x \rightarrow \pi^-} \frac{1}{\sin(x)} = +\infty$$

$$\lim_{x \rightarrow \pi^+} \frac{1}{\sin(x)} = -\infty$$

Example 1.3.12

Again, we can make Definitions [1.3.10](#) and [1.3.11](#) into mathematically precise formal definitions using techniques very similar to those in the optional Section [1.7](#). This is not strictly necessary for this course.

Up to this point we explored limits by sketching graphs or plugging values into a calculator. This was done to help build intuition, but it is not really the basis of a systematic method for computing limits. We have also avoided more formal approaches¹² since we do not have time in the course to go into that level of detail and (arguably) we don't need that detail to achieve the aims of the course. Thankfully we can develop a more systematic approach based on the idea of building up complicated limits from simpler ones by examining how limits interact with the basic operations of arithmetic.

1.4 ▲ Calculating Limits with Limit Laws

Think back to the functions you know and the sorts of things you have been asked to draw, factor and so on. Then they are all constructed from simple pieces, such as

12 The formal approaches are typically referred to as “epsilon-delta limits” or “epsilon-delta proofs” since the symbols ϵ and δ are traditionally used throughout. Take a peek at Section [1.7](#) to see.

- constants — c
- monomials — x^n
- trigonometric functions — $\sin(x)$, $\cos(x)$ and $\tan(x)$

These are the building blocks from which we construct functions. Soon we will add a few more functions to this list, especially the exponential function and various inverse functions.

We then take these building blocks and piece them together using arithmetic

- addition and subtraction — $f(x) = g(x) + h(x)$ and $f(x) = g(x) - h(x)$
- multiplication — $f(x) = g(x) \cdot h(x)$
- division — $f(x) = \frac{g(x)}{h(x)}$
- substitution — $f(x) = g(h(x))$ — this is also called the composition of g with h .

The idea of building up complicated functions from simpler pieces was discussed in Section 0.5.

What we will learn in this section is how to compute the limits of the basic building blocks and then how we can compute limits of sums, products and so forth using “limit laws”. This process allows us to compute limits of complicated functions, using very simple tools and without having to resort to “plugging in numbers” or “closer and closer” or “ $\epsilon - \delta$ arguments”.

In the examples we saw above, almost all the *interesting* limits happened at points where the underlying function was badly behaved — where it jumped, was not defined or blew up to infinity. In those cases we had to be careful and think about what was happening. Thankfully most functions we will see do not have too many points at which these sorts of things happen.

For example, polynomials do not have any nasty jumps and are defined everywhere and do not “blow up”. If you plot them, they look smooth¹³. Polynomials and limits behave very nicely together, and for any polynomial $P(x)$ and any real number a we have that

$$\lim_{x \rightarrow a} P(x) = P(a)$$

That is — to evaluate the limit we just plug in the number. We will build up to this result over the next few pages.

Let us start with the two easiest limits¹⁴.

Theorem 1.4.1 (Easiest limits).

Let $a, c \in \mathbb{R}$. The following two limits hold

$$\lim_{x \rightarrow a} c = c$$

and

$$\lim_{x \rightarrow a} x = a.$$

¹³ We have used this term in an imprecise way, but it does have a precise mathematical meaning.

¹⁴ Though it lies outside the scope of the course, you can find the formal $\epsilon - \delta$ proof of this result at the end of Section 1.7.

Since we have not seen too many theorems yet, let us examine it carefully piece by piece.

- **Let $a, c \in \mathbb{R}$** — just as was the case for definitions, we start a theorem by defining terms and setting the scene. There is not too much scene to set: the symbols a and c are real numbers.
- **The following two limits hold** — this doesn't really contribute much to the statement of the theorem, it just makes it easier to read.
- $\lim_{x \rightarrow a} c = c$ — when we take the limit of a constant function (for example think of $c = 3$), the limit is (unsurprisingly) just that same constant.
- $\lim_{x \rightarrow a} x = a$ — as we noted above for general polynomials, the limit of the function $f(x) = x$ as x approaches a given point a , is just a . This says something quite obvious — as x approaches a , x approaches a (if you are not convinced then sketch the graph).

Armed with only these two limits, we cannot do very much. But combining these limits with some arithmetic we can do quite a lot. For a moment, take a step back from limits for a moment and think about how we construct functions. To make the discussion a little more precise think about how we might construct the function

$$h(x) = \frac{2x - 3}{x^2 + 5x - 6}$$

If we want to compute the value of the function at $x = 2$, then we would

- compute the numerator at $x = 2$
- compute the denominator at $x = 2$
- compute the ratio

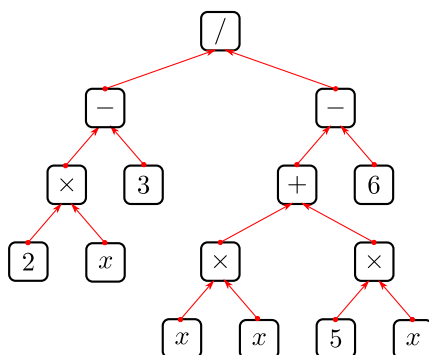
Now to compute the numerator we

- take x and multiply it by 2
- subtract 3 to the result

While for the denominator

- multiply x by x
- multiply x by 5
- add these two numbers and subtract 6

This sequence of operations can be represented pictorially as the tree shown in Figure 1.4.1 below.

Figure 1.4.1.

Such trees were discussed in Section 0.5 (now is not a bad time to quickly review that section before proceeding). The point here is that in order to compute the value of the function we just repeatedly add, subtract, multiply and divide constants and x .

To compute the limit of the above function at $x = 2$ we can do something very similar. From the previous theorem we know how to compute

$$\lim_{x \rightarrow 2} c = c$$

and

$$\lim_{x \rightarrow 2} x = 2$$

and the next theorem will tell us how to stitch together these two limits using the arithmetic we used to construct the function.

Theorem 1.4.2 (Arithmetic of limits).

Let $a, c \in \mathbb{R}$, let $f(x)$ and $g(x)$ be defined for all x 's that lie in some interval about a (but f, g need not be defined exactly at a).

$$\lim_{x \rightarrow a} f(x) = F$$

$$\lim_{x \rightarrow a} g(x) = G$$

exist with $F, G \in \mathbb{R}$. Then the following limits hold

- $\lim_{x \rightarrow a} (f(x) + g(x)) = F + G$ — limit of the sum is the sum of the limits.
- $\lim_{x \rightarrow a} (f(x) - g(x)) = F - G$ — limit of the difference is the difference of the limits.
- $\lim_{x \rightarrow a} cf(x) = cF$.
- $\lim_{x \rightarrow a} (f(x) \cdot g(x)) = F \cdot G$ — limit of the product is the product of limits.
- If $G \neq 0$ then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{F}{G}$ and, in particular, $\lim_{x \rightarrow a} \frac{1}{g(x)} = \frac{1}{G}$.

Note — be careful with this last one — the denominator cannot be zero.

The above theorem shows that limits interact very simply with arithmetic. If you are asked to find the limit of a sum then the answer is just the sum of the limits. Similarly the limit of a product is just the product of the limits.

How do we apply the above theorem to the rational function $h(x)$ we defined above? Here is a warm-up example:

Example 1.4.3

You are given two functions f, g (not explicitly) which have the following limits as x approaches 1:

$$\lim_{x \rightarrow 1} f(x) = 3 \quad \text{and} \quad \lim_{x \rightarrow 1} g(x) = 2$$

Using the above theorem we can compute

$$\begin{aligned} \lim_{x \rightarrow 1} 3f(x) &= 3 \times 3 = 9 \\ \lim_{x \rightarrow 1} 3f(x) - g(x) &= 3 \times 3 - 2 = 7 \\ \lim_{x \rightarrow 1} f(x)g(x) &= 3 \times 2 = 6 \\ \lim_{x \rightarrow 1} \frac{f(x)}{f(x) - g(x)} &= \frac{3}{3 - 2} = 3 \end{aligned}$$

Example 1.4.3

Another simple example

Example 1.4.4

Find $\lim_{x \rightarrow 3} 4x^2 - 1$

We use the arithmetic of limits:

$$\begin{aligned} \lim_{x \rightarrow 3} 4x^2 - 1 &= \left(\lim_{x \rightarrow 3} 4x^2 \right) - \lim_{x \rightarrow 3} 1 && \text{difference of limits} \\ &= \left(\lim_{x \rightarrow 3} 4 \cdot \lim_{x \rightarrow 3} x^2 \right) - \lim_{x \rightarrow 3} 1 && \text{product of limits} \\ &= 4 \cdot \left(\lim_{x \rightarrow 3} x^2 \right) - 1 && \text{limit of constant} \\ &= 4 \cdot \left(\lim_{x \rightarrow 3} x \right) \cdot \left(\lim_{x \rightarrow 3} x \right) - 1 && \text{product of limits} \\ &= 4 \cdot 3 \cdot 3 - 1 && \text{limit of } x \\ &= 36 - 1 \\ &= 35 \end{aligned}$$

Example 1.4.4

This is an excruciating level of detail, but when you first use this theorem and try some

examples it is a good idea to do things step by step by step until you are comfortable with it.

Example 1.4.5

Yet another limit — compute $\lim_{x \rightarrow 2} \frac{x}{x-1}$.

To apply the arithmetic of limits, we need to examine numerator and denominator separately and make sure the limit of the denominator is non-zero. Numerator first:

$$\lim_{x \rightarrow 2} x = 2 \quad \text{limit of } x$$

and now the denominator:

$$\begin{aligned} \lim_{x \rightarrow 2} x - 1 &= \left(\lim_{x \rightarrow 2} x \right) - \left(\lim_{x \rightarrow 2} 1 \right) && \text{difference of limits} \\ &= 2 - 1 && \text{limit of } x \text{ and limit of constant} = 1 \end{aligned}$$

Since the limit of the denominator is non-zero we can put it back together to get

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x}{x-1} &= \frac{\lim_{x \rightarrow 2} x}{\lim_{x \rightarrow 2} (x-1)} \\ &= \frac{2}{1} \\ &= 2 \end{aligned}$$

Example 1.4.5

In the next example we show that many different things can happen if the limit of the denominator is zero.

Example 1.4.6 (Be careful with limits of ratios)

We must be careful when computing the limit of a ratio — it is the ratio of the limits except when the limit of the denominator is zero. When the limit of the denominator is zero Theorem [1.4.2](#) **does not apply** and a few interesting things can happen

- If the limit of the numerator is non-zero then the limit of the ratio does not exist

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = DNE \quad \text{when } \lim_{x \rightarrow a} f(x) \neq 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

For example, $\lim_{x \rightarrow 0} \frac{1}{x^2} = DNE$.

- If the limit of the numerator is zero then the above theorem does not give us enough information to decide whether or not the limit exists. It is possible that

$$\text{– the limit does not exist, eg. } \lim_{x \rightarrow 0} \frac{x}{x^2} = \lim_{x \rightarrow 0} \frac{1}{x} = DNE$$

- the limit is $\pm\infty$, eg. $\lim_{x \rightarrow 0} \frac{x^2}{x^4} = \lim_{x \rightarrow 0} \frac{1}{x^2} = +\infty$ or $\lim_{x \rightarrow 0} \frac{-x^2}{x^4} = \lim_{x \rightarrow 0} \frac{-1}{x^2} = -\infty$.
- the limit is zero, eg. $\lim_{x \rightarrow 0} \frac{x^2}{x} = 0$
- the limit exists and is non-zero, eg. $\lim_{x \rightarrow 0} \frac{x}{x} = 1$

Now while the above examples are very simple and a little contrived they serve to illustrate the point we are trying to make — be careful if the limit of the denominator is zero.

Example 1.4.6

We now have enough theory to return to our rational function and compute its limit as x approaches 2.

Example 1.4.7

Let $h(x) = \frac{2x - 3}{x^2 + 5x - 6}$ and find its limit as x approaches 2.

Since this is the limit of a ratio, we compute the limit of the numerator and denominator separately. Numerator first:

$$\begin{aligned}
 \lim_{x \rightarrow 2} 2x - 3 &= \left(\lim_{x \rightarrow 2} 2x \right) - \left(\lim_{x \rightarrow 2} 3 \right) && \text{difference of limits} \\
 &= 2 \cdot \left(\lim_{x \rightarrow 2} x \right) - 3 && \text{product of limits and limit of constant} \\
 &= 2 \cdot 2 - 3 && \text{limits of } x \\
 &= 1
 \end{aligned}$$

Denominator next:

$$\begin{aligned}
 \lim_{x \rightarrow 2} x^2 + 5x - 6 &= \left(\lim_{x \rightarrow 2} x^2 \right) + \left(\lim_{x \rightarrow 2} 5x \right) - \left(\lim_{x \rightarrow 2} 6 \right) && \text{sum of limits} \\
 &= \left(\lim_{x \rightarrow 2} x \right) \cdot \left(\lim_{x \rightarrow 2} x \right) + 5 \cdot \left(\lim_{x \rightarrow 2} x \right) - 6 && \text{product of limits and limit of constant} \\
 &= 2 \cdot 2 + 5 \cdot 2 - 6 && \text{limits of } x \\
 &= 8
 \end{aligned}$$

Since the limit of the denominator is non-zero, we can obtain our result by taking the ratio of the separate limits.

$$\lim_{x \rightarrow 2} \frac{2x - 3}{x^2 + 5x - 6} = \frac{\lim_{x \rightarrow 2} 2x - 3}{\lim_{x \rightarrow 2} x^2 + 5x - 6} = \frac{1}{8}$$

The above works out quite simply. However, if we were to take the limit as $x \rightarrow 1$ then things are a bit harder. The limit of the numerator is:

$$\lim_{x \rightarrow 1} 2x - 3 = 2 \cdot 1 - 3 = -1$$

(we have not listed all the steps). And the limit of the denominator is

$$\lim_{x \rightarrow 1} x^2 + 5x - 6 = 1 \cdot 1 + 5 - 6 = 0$$

Since the limit of the numerator is non-zero, while the limit of the denominator is zero, the limit of the ratio does not exist.

$$\lim_{x \rightarrow 1} \frac{2x - 3}{x^2 + 5x - 6} = DNE$$

Example 1.4.7

It is **IMPORTANT TO NOTE** that it is not correct to write

$$\lim_{x \rightarrow 1} \frac{2x - 3}{x^2 + 5x - 6} = \frac{-1}{0} = DNE$$

Because we can only write

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \text{something}$$

when the limit of the denominator is non-zero (see Example 1.4.6 above).

With a little care you can use the arithmetic of limits to obtain the following rules for limits of powers of functions and limits of roots of functions:

Theorem 1.4.8 (More arithmetic of limits — powers and roots).

Let n be a positive integer, let $a \in \mathbb{R}$ and let f be a function so that

$$\lim_{x \rightarrow a} f(x) = F$$

for some real number F . Then the following holds

$$\lim_{x \rightarrow a} (f(x))^n = \left(\lim_{x \rightarrow a} f(x) \right)^n = F^n$$

so that the limit of a power is the power of the limit. Similarly, if

- n is an even number and $F > 0$, or
- n is an odd number and F is any real number

then

$$\lim_{x \rightarrow a} (f(x))^{1/n} = \left(\lim_{x \rightarrow a} f(x) \right)^{1/n} = F^{1/n}$$

More generally¹⁵, if $F > 0$ and p is any real number,

$$\lim_{x \rightarrow a} (f(x))^p = \left(\lim_{x \rightarrow a} f(x) \right)^p = F^p$$

Notice that we have to be careful when taking roots of limits that might be negative numbers. To see why, consider the case $n = 2$, the limit

$$\begin{aligned}\lim_{x \rightarrow 4} x^{1/2} &= 4^{1/2} = 2 \\ \lim_{x \rightarrow 4} (-x)^{1/2} &= (-4)^{1/2} = \text{not a real number}\end{aligned}$$

In order to evaluate such limits properly we need to use complex numbers which are beyond the scope of this text.

Also note that the notation $x^{1/2}$ refers to the *positive* square root of x . While 2 and (-2) are both square-roots of 4, the notation $4^{1/2}$ means 2. This is something we must be careful of¹⁶

So again — let us do a few examples and carefully note what we are doing.

Example 1.4.9

$$\begin{aligned}\lim_{x \rightarrow 2} (4x^2 - 3)^{1/3} &= \left(\left(\lim_{x \rightarrow 2} 4x^2 \right) - \left(\lim_{x \rightarrow 2} 3 \right) \right)^{1/3} \\ &= \left(4 \cdot 2^2 - 3 \right)^{1/3} \\ &= (16 - 3)^{1/3} \\ &= 13^{1/3}\end{aligned}$$

Example 1.4.9

By combining the last few theorems we can make the evaluation of limits of polynomials and rational functions much easier:

Theorem 1.4.10 (Limits of polynomials and rational functions).

Let $a \in \mathbb{R}$, let $P(x)$ be a polynomial and let $R(x)$ be a rational function. Then

$$\lim_{x \rightarrow a} P(x) = P(a)$$

and provided $R(x)$ is defined at $x = a$ then

$$\lim_{x \rightarrow a} R(x) = R(a)$$

If $R(x)$ is not defined at $x = a$ then we are not able to apply this result.

- 15 You may not know the definition of the power b^p when p is not a rational number, so here it is. If $b > 0$ and p is any real number, then b^p is the limit of b^r as r approaches p through rational numbers. We won't do so here, but it is possible to prove that the limit exists.
- 16 Like ending sentences in prepositions — “This is something up with which we will not put.” This quote is attributed to Churchill though there is some dispute as to whether or not he really said it.

So the previous examples are now much easier to compute:

$$\lim_{x \rightarrow 2} \frac{2x - 3}{x^2 + 5x - 6} = \frac{4 - 3}{4 + 10 - 6} = \frac{1}{8}$$

$$\lim_{x \rightarrow 2} (4x^2 - 1) = 16 - 1 = 15$$

$$\lim_{x \rightarrow 2} \frac{x}{x - 1} = \frac{2}{2 - 1} = 2$$

It is clear that limits of polynomials are very easy, while those of rational functions are easy except when the denominator might go to zero. We have seen examples where the resulting limit does not exist, and some where it does. We now work to explain this more systematically. The following example demonstrates that it is sometimes possible to take the limit of a rational function to a point at which the denominator is zero. Indeed we must be able to do exactly this in order to be able to define derivatives in the next chapter.

Example 1.4.11

Consider the limit

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2}{x - 1}.$$

If we try to apply the arithmetic of limits then we compute the limits of the numerator and denominator separately

$$\lim_{x \rightarrow 1} x^3 - x^2 = 1 - 1 = 0 \quad (1.4.1)$$

$$\lim_{x \rightarrow 1} x - 1 = 1 - 1 = 0 \quad (1.4.2)$$

Since the denominator is zero, we cannot apply our theorem and we are, for the moment, stuck. However, there is more that we can do here — the hint is that the numerator and denominator *both* approach zero as x approaches 1. This means that there might be something we can cancel.

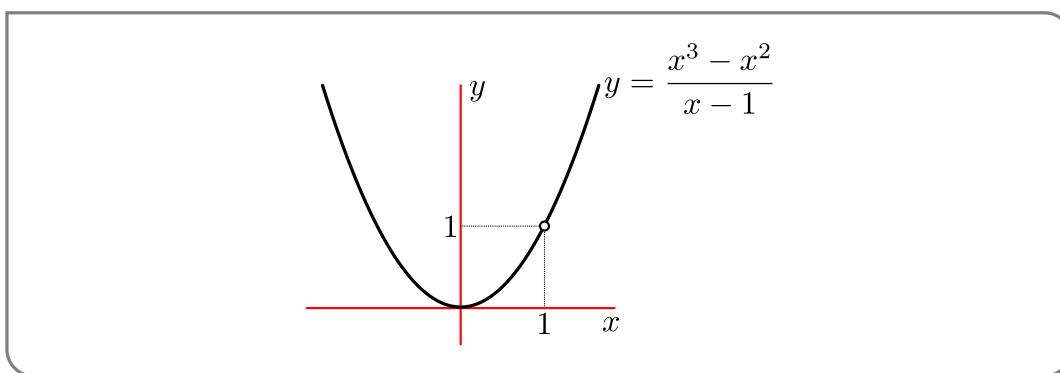
So let us play with the expression a little more before we take the limit:

$$\frac{x^3 - x^2}{x - 1} = \frac{x^2(x - 1)}{x - 1} = x^2 \quad \text{provided } x \neq 1.$$

So what we really have here is the following function

$$\frac{x^3 - x^2}{x - 1} = \begin{cases} x^2 & x \neq 1 \\ \text{undefined} & x = 1 \end{cases}$$

If we plot the above function the graph looks exactly the same as $y = x^2$ except that the function is not defined at $x = 1$ (since at $x = 1$ both numerator and denominator are zero).



When we compute a limit as $x \rightarrow a$, the value of the function exactly at $x = a$ is irrelevant. We only care what happens to the function as we bring x very close to a . So for the above problem we can write

$$\frac{x^3 - x^2}{x - 1} = x^2 \quad \text{when } x \text{ is close to } 1 \text{ but not at } x = 1$$

So the limit as $x \rightarrow 1$ of the function is the same as the limit $\lim_{x \rightarrow 1} x^2$ since the functions are the same except exactly at $x = 1$. By this reasoning we get

$$\lim_{x \rightarrow 1} \frac{x^3 - x^2}{x - 1} = \lim_{x \rightarrow 1} x^2 = 1$$

Example 1.4.11

The reasoning in the above example can be made more general:

Theorem 1.4.12.

If $f(x) = g(x)$ except when $x = a$ then $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x)$ provided the limit of g exists.

How do we know when to use this theorem? The big clue is that when we try to compute the limit in a naive way, we end up with $\frac{0}{0}$. We know that $\frac{0}{0}$ does not make sense, but it is an indication that there might be a common factor between numerator and denominator that can be cancelled. In the previous example, this common factor was $(x - 1)$.

Example 1.4.13

Using this idea compute

$$\lim_{h \rightarrow 0} \frac{(1 + h)^2 - 1}{h}$$

- First we should check that we cannot just substitute $h = 0$ into this — clearly we cannot because the denominator would be 0.

- But we should also check the numerator to see if we have $\frac{0}{0}$, and we see that the numerator gives us $1 - 1 = 0$.
- Thus we have a hint that there is a common factor that we might be able to cancel. So now we look for the common factor and try to cancel it.

$$\begin{aligned}\frac{(1+h)^2 - 1}{h} &= \frac{1 + 2h + h^2 - 1}{h} && \text{expand} \\ &= \frac{2h + h^2}{h} = \frac{h(2+h)}{h} && \text{factor and then cancel} \\ &= 2 + h\end{aligned}$$

- Thus we really have that

$$\frac{(1+h)^2 - 1}{h} = \begin{cases} 2 + h & h \neq 0 \\ \text{undefined} & h = 0 \end{cases}$$

and because of this

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} &= \lim_{h \rightarrow 0} 2 + h \\ &= 2\end{aligned}$$

Example 1.4.13

Of course — we have written everything out in great detail here and that is way more than is required for a solution to such a problem. Let us do it again a little more succinctly.

Example 1.4.14

Compute the following limit:

$$\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h}$$

If we try to use the arithmetic of limits, then we see that the limit of the numerator and the limit of the denominator are both zero. Hence we should try to factor them and cancel any common factor. This gives

$$\begin{aligned}\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} &= \lim_{h \rightarrow 0} \frac{1 + 2h + h^2 - 1}{h} \\ &= \lim_{h \rightarrow 0} 2 + h \\ &= 2\end{aligned}$$

Example 1.4.14

Notice that even though we did this example carefully above, we have still written some text in our working explaining what we have done. You should always think about the

reader and if in doubt, put in more explanation rather than less. We could make the above example even more terse

Example 1.4.15

Compute the following limit:

$$\lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h}$$

Numerator and denominator both go to zero as $h \rightarrow 0$. So factor and simplify:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{(1+h)^2 - 1}{h} &= \lim_{h \rightarrow 0} \frac{1 + 2h + h^2 - 1}{h} \\ &= \lim_{h \rightarrow 0} 2 + h = 2 \end{aligned}$$

Example 1.4.15

A slightly harder one now

Example 1.4.16

Compute the limit

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x} - 1}$$

If we try to use the arithmetic of limits we get

$$\begin{aligned} \lim_{x \rightarrow 0} x &= 0 \\ \lim_{x \rightarrow 0} \sqrt{1+x} - 1 &= \sqrt{\lim_{x \rightarrow 0} 1+x} - 1 = 1 - 1 = 0 \end{aligned}$$

So doing the naive thing we'd get $0/0$. This suggests a common factor that can be cancelled. Since the numerator and denominator are not polynomials we have to try other tricks¹⁷. We can simplify the denominator $\sqrt{1+x} - 1$ a lot, and in particular eliminate

17 While these tricks are useful (and even cute¹⁸), Taylor polynomials (see Section 3.4) give us a more systematic way of approaching this problem.

18 Mathematicians tend to have quite strong opinions on the beauty of mathematics. For example, Paul Erdős¹⁹ said “Why are numbers beautiful? It’s like asking why is Beethoven’s Ninth Symphony beautiful. If you don’t see why, someone can’t tell you. I know numbers are beautiful. If they aren’t beautiful, nothing is.”.

19 Arguably the most prolific mathematician of the 20th century — definitely worth a google. The authors do not know his opinion on nested footnotes²⁰.

20 Nested footnotes are generally frowned upon, since they can get quite contorted; see XKCD-1208 and also the novel “House of Leaves” by Mark Z. Danielewski.

the square root, by multiplying it by its conjugate $\sqrt{1+x}+1$.

$$\begin{aligned}
 \frac{x}{\sqrt{1+x}-1} &= \frac{x}{\sqrt{1+x}-1} \times \frac{\sqrt{1+x}+1}{\sqrt{1+x}+1} && \text{multiply by } \frac{\text{conjugate}}{\text{conjugate}} = 1 \\
 &= \frac{x(\sqrt{1+x}+1)}{(\sqrt{1+x}-1)(\sqrt{1+x}+1)} && \text{bring things together} \\
 &= \frac{x(\sqrt{1+x}+1)}{(\sqrt{1+x})^2 - 1 \cdot 1} && \text{since } (a-b)(a+b) = a^2 - b^2 \\
 &= \frac{x(\sqrt{1+x}+1)}{1+x-1} && \text{clean up a little} \\
 &= \frac{x(\sqrt{1+x}+1)}{x} \\
 &= \sqrt{1+x}+1 && \text{cancel the } x
 \end{aligned}$$

So now we have

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{x}{\sqrt{1+x}-1} &= \lim_{x \rightarrow 0} \sqrt{1+x}+1 \\
 &= \sqrt{1+0}+1 = 2
 \end{aligned}$$

Example 1.4.16

How did we know what to multiply by? Our function was of the form

$$\frac{a}{\sqrt{b}-c}$$

so, to eliminate the square root from the denominator, we employ a trick — we multiply by 1. Of course, multiplying by 1 doesn't do anything. But if you multiply by 1 carefully you can leave the value the same, but change the form of the expression. More precisely

$$\begin{aligned}
 \frac{a}{\sqrt{b}-c} &= \frac{a}{\sqrt{b}-c} \cdot 1 \\
 &= \frac{a}{\sqrt{b}-c} \cdot \underbrace{\frac{\sqrt{b}+c}{\sqrt{b}+c}}_{=1} \\
 &= \frac{a(\sqrt{b}+c)}{(\sqrt{b}-c)(\sqrt{b}+c)} && \text{expand denominator carefully} \\
 &= \frac{a(\sqrt{b}+c)}{\sqrt{b} \cdot \sqrt{b} - c\sqrt{b} + c\sqrt{b} - c \cdot c} && \text{do some cancellation} \\
 &= \frac{a(\sqrt{b}+c)}{b-c^2}
 \end{aligned}$$

Now the numerator contains roots, but the denominator is just a polynomial.

Before we move on to limits at infinity, there is one more theorem to see. While the scope of its application is quite limited, it can be extremely useful. It is called a sandwich theorem or a squeeze theorem for reasons that will become apparent.

Sometimes one is presented with an unpleasant ugly function such as

$$f(x) = x^2 \sin(\pi/x)$$

It is a fact of life, that not all the functions that are encountered in mathematics will be elegant and simple; this is especially true when the mathematics gets applied to real world problems. One just has to work with what one gets. So how can we compute

$$\lim_{x \rightarrow 0} x^2 \sin(\pi/x)?$$

Since it is the product of two functions, we might try

$$\begin{aligned} \lim_{x \rightarrow 0} x^2 \sin(\pi/x) &= \left(\lim_{x \rightarrow 0} x^2 \right) \cdot \left(\lim_{x \rightarrow 0} \sin(\pi/x) \right) \\ &= 0 \cdot \left(\lim_{x \rightarrow 0} \sin(\pi/x) \right) \\ &= 0 \end{aligned}$$

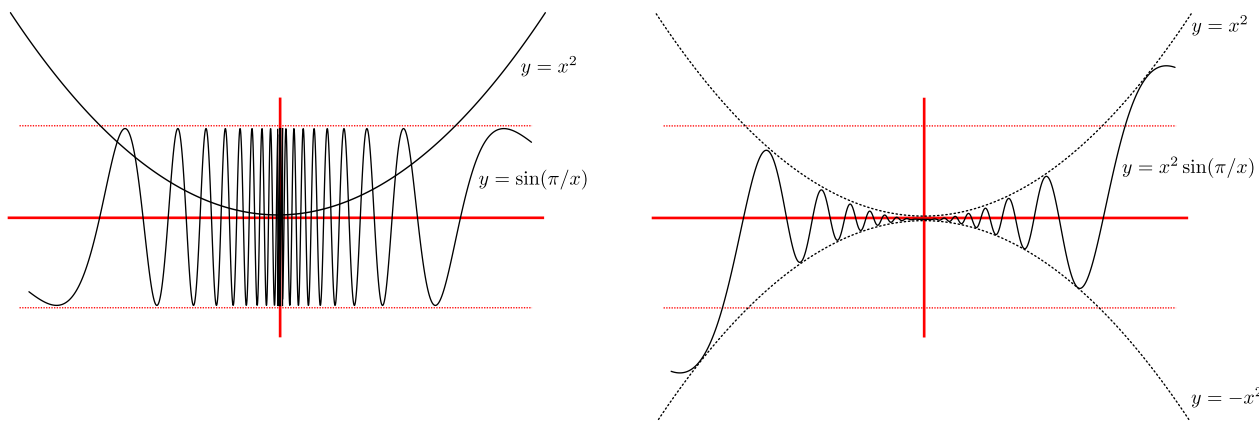
But we just cheated — we cannot use the arithmetic of limits theorem here, because the limit

$$\lim_{x \rightarrow 0} \sin(\pi/x) = DNE$$

does not exist. Now we did see the function $\sin(\pi/x)$ before (in Example [1.3.5](#)), so you should go back and look at it again. Unfortunately the theorem “the limit of a product is the product of the limits” only holds when the limits you are trying to multiply together actually exist. So we cannot use it.

However, we do see that the function naturally decomposes into the product of two pieces — the functions x^2 and $\sin(\pi/x)$. We have sketched the two functions in the figure on the left below.

Figure 1.4.2.



While x^2 is a very well behaved function and we know quite a lot about it, the function $\sin(\pi/x)$ is quite ugly. One of the few things we can say about it is the following

$$-1 \leq \sin(\pi/x) \leq 1 \quad \text{provided } x \neq 0$$

But if we multiply this expression by x^2 we get (because $x^2 \geq 0$)

$$-x^2 \leq x^2 \sin(\pi/x) \leq x^2 \quad \text{provided } x \neq 0$$

and we have sketched the result in the figure above (on the right). So the function we are interested in is *squeezed* or *sandwiched* between the functions x^2 and $-x^2$.

If we focus in on the picture close to $x = 0$ we see that x approaches 0, the functions x^2 and $-x^2$ both approach 0. Further, because $x^2 \sin(\pi/x)$ is sandwiched between them, it seems that it also approaches 0.

The following theorem tells us that this is indeed the case:

Theorem 1.4.17 (Squeeze theorem (or sandwich theorem or pinch theorem)).

Let $a \in \mathbb{R}$ and let f, g, h be three functions so that

$$f(x) \leq g(x) \leq h(x)$$

for all x in an interval around a , except possibly exactly at $x = a$. Then if

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then it is also the case that

$$\lim_{x \rightarrow a} g(x) = L$$

Using the above theorem we can compute the limit we want and write it up nicely

Example 1.4.18

Compute the limit

$$\lim_{x \rightarrow 0} x^2 \sin(\pi/x)$$

Since $-1 \leq \sin(\theta) \leq 1$ for all real numbers θ , we have

$$-1 \leq \sin(\pi/x) \leq 1 \quad \text{for all } x \neq 0$$

Multiplying the above by x^2 we see that

$$-x^2 \leq x^2 \sin(\pi/x) \leq x^2 \quad \text{for all } x \neq 0$$

Since $\lim_{x \rightarrow 0} x^2 = \lim_{x \rightarrow 0} (-x^2) = 0$ by the sandwich (or squeeze or pinch) theorem we have

$$\lim_{x \rightarrow 0} x^2 \sin(\pi/x) = 0$$

Example 1.4.18

Notice how we have used “words”. We have remarked on this several times already in the text, but we will keep mentioning it. It is okay to use words in your answers to maths problems — and you should do so! These let the reader know what you are doing and help you understand what you are doing.

Another sandwich theorem example

Example 1.4.19

Let $f(x)$ be a function such that $1 \leq f(x) \leq x^2 - 2x + 2$. What is $\lim_{x \rightarrow 1} f(x)$?

We are already supplied with an inequality, so it is likely that it is going to help us. We should examine the limits of each side to see if they are the same:

$$\begin{aligned}\lim_{x \rightarrow 1} 1 &= 1 \\ \lim_{x \rightarrow 1} x^2 - 2x + 2 &= 1 - 2 + 2 = 1\end{aligned}$$

So we see that the function $f(x)$ is trapped between two functions that both approach 1 as $x \rightarrow 1$. Hence by the sandwich / pinch / squeeze theorem, we know that

$$\lim_{x \rightarrow 1} f(x) = 1$$

Example 1.4.19

To get some intuition as to why the squeeze theorem is true, consider when x is very very close to a . In particular, consider when x is sufficiently close to a that we know $h(x)$ is within 10^{-6} of L and that $f(x)$ is also within 10^{-6} of L . That is

$$|h(x) - L| < 10^{-6} \quad \text{and} \quad |f(x) - L| < 10^{-6}.$$

This means that

$$L - 10^{-6} < f(x) \leq h(x) < L + 10^{-6}$$

since we know that $f(x) \leq h(x)$.

But now by the hypothesis of the squeeze theorem we know that $f(x) \leq g(x) \leq h(x)$ and so we have

$$L - 10^{-6} < f(x) \leq g(x) \leq h(x) < L + 10^{-6}$$

And thus we know that

$$L - 10^{-6} \leq g(x) \leq L + 10^{-6} \tag{1.4.3}$$

That is $g(x)$ is also within 10^{-6} of L .

In this argument our choice of 10^{-6} was arbitrary, so we can really replace 10^{-6} with any small number we like. Hence we know that we can force $g(x)$ as close to L as we like, by bringing x sufficiently close to a . We give a more formal and rigorous version of this argument at the end of Section [1.9](#).

- (a) What is the average velocity of the object from 3 seconds to 5 seconds?
- (b) What is the velocity of the object at time $t = 1$?

Q[8]: Suppose the position of a body at time t (measured in seconds) is given by $s(t) = \sqrt{t}$.

- (a) What is the average velocity of the object from $t = 1$ second to $t = 9$ seconds?
- (b) What is the velocity of the object at time $t = 1$?
- (c) What is the velocity of the object at time $t = 9$?

1.3▲ The Limit of a Function

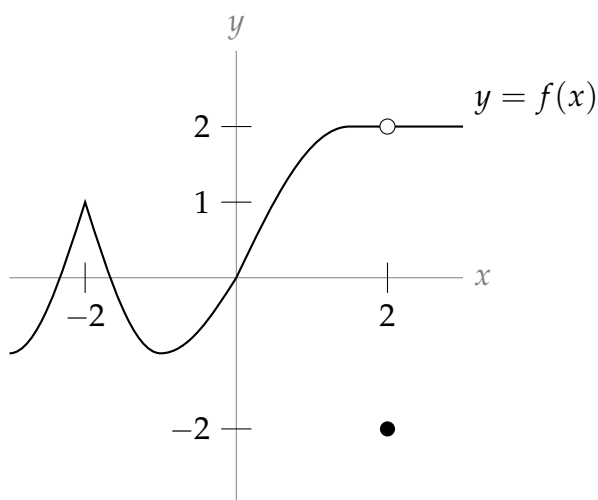
Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

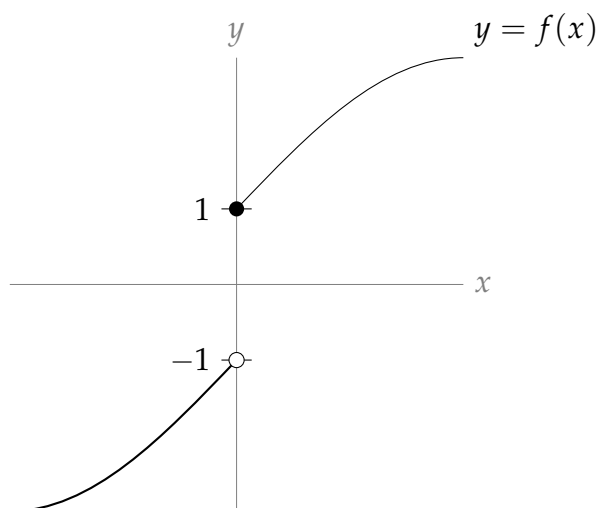
► Stage 1

Q[1]: Given the function shown below, evaluate the following:

- (a) $\lim_{x \rightarrow -2} f(x)$
- (b) $\lim_{x \rightarrow 0} f(x)$
- (c) $\lim_{x \rightarrow 2} f(x)$



Q[2]: Given the function shown below, evaluate $\lim_{x \rightarrow 0} f(x)$.



Q[3]: Given the function shown below, evaluate:

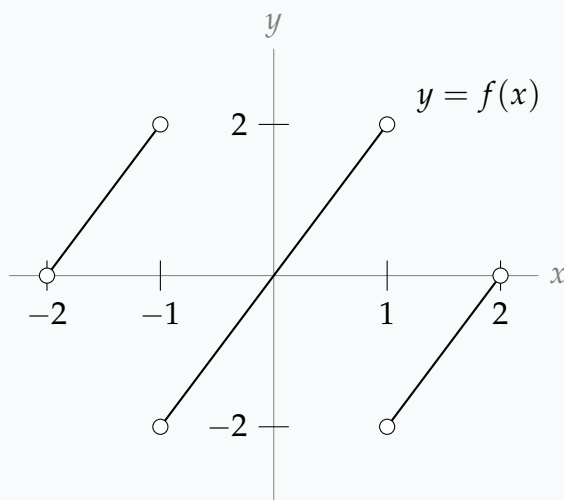
(a) $\lim_{x \rightarrow -1^-} f(x)$

(b) $\lim_{x \rightarrow -1^+} f(x)$

(c) $\lim_{x \rightarrow -1} f(x)$

(d) $\lim_{x \rightarrow -2^+} f(x)$

(e) $\lim_{x \rightarrow 2^-} f(x)$



Q[4]: Draw a curve $y = f(x)$ with $\lim_{x \rightarrow 3} f(x) = f(3) = 10$.

Q[5]: Draw a curve $y = f(x)$ with $\lim_{x \rightarrow 3} f(x) = 10$ and $f(3) = 0$.

Q[6]: Suppose $\lim_{x \rightarrow 3} f(x) = 10$. True or false: $f(3) = 10$.

Q[7]: Suppose $f(3) = 10$. True or false: $\lim_{x \rightarrow 3} f(x) = 10$.

Q[8]: Suppose $f(x)$ is a function defined on all real numbers, and $\lim_{x \rightarrow -2} f(x) = 16$. What is $\lim_{x \rightarrow -2^-} f(x)$?

Q[9]: Suppose $f(x)$ is a function defined on all real numbers, and $\lim_{x \rightarrow -2^-} f(x) = 16$. What is $\lim_{x \rightarrow -2} f(x)$?

► Stage 2

In Questions [10](#) through [17](#), evaluate the given limits. If you aren't sure where to begin, it's nice to start by drawing the function.

Q[10]: $\lim_{t \rightarrow 0} \sin t$

Q[11]: $\lim_{x \rightarrow 0^+} \log x$

Q[12]: $\lim_{y \rightarrow 3} y^2$

Q[13]: $\lim_{x \rightarrow 0^-} \frac{1}{x}$

Q[14]: $\lim_{x \rightarrow 0} \frac{1}{x}$

Q[15]: $\lim_{x \rightarrow 0} \frac{1}{x^2}$

Q[16]: $\lim_{x \rightarrow 3} \frac{1}{10}$

Q[17]: $\lim_{x \rightarrow 3} f(x)$, where $f(x) = \begin{cases} \sin x & x \leq 2.9 \\ x^2 & x > 2.9 \end{cases}$.

1.4▲ Calculating Limits with Limit Laws

Exercises

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►► Stage 1

Q[1]: Suppose $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$. Which of the following limits can you compute, given this information?

- (a) $\lim_{x \rightarrow a} \frac{f(x)}{2}$
- (b) $\lim_{x \rightarrow a} \frac{2}{f(x)}$
- (c) $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$
- (d) $\lim_{x \rightarrow a} f(x)g(x)$

Q[2]: Give two functions $f(x)$ and $g(x)$ that satisfy $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} g(x) = 0$ and $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = 10$.

Q[3]: Give two functions $f(x)$ and $g(x)$ that satisfy $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} g(x) = 0$ and $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = 0$.

Q[4]: Give two functions $f(x)$ and $g(x)$ that satisfy $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} g(x) = 0$ and $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = \infty$.

Q[5]: Suppose $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$. What are the possible values of $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$?

►► Stage 2

For Questions 6 through 41, evaluate the given limits.

Q[6]: $\lim_{t \rightarrow 10} \frac{2(t-10)^2}{t}$

Q[7]: $\lim_{y \rightarrow 0} \frac{(y+1)(y+2)(y+3)}{\cos y}$

Q[8]: $\lim_{x \rightarrow 3} \left(\frac{4x-2}{x+2} \right)^4$

Q[9](*) : $\lim_{t \rightarrow -3} \left(\frac{1-t}{\cos(t)} \right)$

$$\text{Q[10](*)}: \lim_{h \rightarrow 0} \frac{(2+h)^2 - 4}{2h}$$

$$\text{Q[11](*)}: \lim_{t \rightarrow -2} \left(\frac{t-5}{t+4} \right)$$

$$\text{Q[12](*)}: \lim_{x \rightarrow 1} \sqrt{5x^3 + 4}$$

$$\text{Q[13](*)}: \lim_{t \rightarrow -1} \left(\frac{t-2}{t+3} \right)$$

$$\text{Q[14](*)}: \lim_{x \rightarrow 1} \frac{\log(1+x) - x}{x^2}$$

$$\text{Q[15](*)}: \lim_{x \rightarrow 2} \left(\frac{x-2}{x^2-4} \right)$$

$$\text{Q[16](*)}: \lim_{x \rightarrow 4} \frac{x^2 - 4x}{x^2 - 16}$$

$$\text{Q[17](*)}: \lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2}$$

$$\text{Q[18](*)}: \lim_{x \rightarrow -3} \frac{x^2 - 9}{x + 3}$$

$$\text{Q[19]}: \lim_{t \rightarrow 2} \frac{1}{2}t^4 - 3t^3 + t$$

$$\text{Q[20](*)}: \lim_{x \rightarrow -1} \frac{\sqrt{x^2 + 8} - 3}{x + 1}.$$

$$\text{Q[21](*)}: \lim_{x \rightarrow 2} \frac{\sqrt{x+7} - \sqrt{11-x}}{2x-4}.$$

$$\text{Q[22](*)}: \lim_{x \rightarrow 1} \frac{\sqrt{x+2} - \sqrt{4-x}}{x-1}$$

$$\text{Q[23](*)}: \lim_{x \rightarrow 3} \frac{\sqrt{x-2} - \sqrt{4-x}}{x-3}.$$

$$\text{Q[24](*)}: \lim_{t \rightarrow 1} \frac{3t-3}{2-\sqrt{5-t}}.$$

$$\text{Q[25]: } \lim_{x \rightarrow 0} -x^2 \cos\left(\frac{3}{x}\right)$$

$$\text{Q[26]: } \lim_{x \rightarrow 0} \frac{x^4 \sin\left(\frac{1}{x}\right) + 5x^2 \cos\left(\frac{1}{x}\right) + 2}{(x-2)^2}$$

$$\text{Q[27](*)}: \lim_{x \rightarrow 0} x \sin^2\left(\frac{1}{x}\right)$$

$$\text{Q[28]: } \lim_{w \rightarrow 5} \frac{2w^2 - 50}{(w-5)(w-1)}$$

$$\text{Q[29]: } \lim_{r \rightarrow -5} \frac{r}{r^2 + 10r + 25}$$

$$\text{Q[30]: } \lim_{x \rightarrow -1} \sqrt{\frac{x^3 + x^2 + x + 1}{3x + 3}}$$

$$\text{Q[31]: } \lim_{x \rightarrow 0} \frac{x^2 + 2x + 1}{3x^5 - 5x^3}$$

$$\text{Q[32]: } \lim_{t \rightarrow 7} \frac{t^2 x^2 + 2tx + 1}{t^2 - 14t + 49}, \text{ where } x \text{ is a positive constant}$$

$$\text{Q[33]: } \lim_{d \rightarrow 0} x^5 - 32x + 15, \text{ where } x \text{ is a constant}$$

$$\text{Q[34]: } \lim_{x \rightarrow 1} (x-1)^2 \sin\left[\left(\frac{x^2 - 3x + 2}{x^2 - 2x + 1}\right)^2 + 15\right]$$

Q[35](*): Evaluate

$$\lim_{x \rightarrow 0} x^{1/101} \sin(x^{-100})$$

or explain why this limit does not exist.

$$\text{Q[36](*)}: \lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x}$$

$$\text{Q[37]: } \lim_{x \rightarrow 5} \frac{(x-5)^2}{x+5}$$

$$\text{Q[38]: Evaluate } \lim_{t \rightarrow \frac{1}{2}} \frac{\frac{1}{3t^2} + \frac{1}{t^2-1}}{2t-1}.$$

$$\text{Q[39]: Evaluate } \lim_{x \rightarrow 0} \left(3 + \frac{|x|}{x}\right).$$

Q[40]: Evaluate $\lim_{d \rightarrow -4} \frac{|3d + 12|}{d + 4}$

Q[41]: Evaluate $\lim_{x \rightarrow 0} \frac{5x - 9}{|x| + 2}$.

Q[42]: Suppose $\lim_{x \rightarrow -1} f(x) = -1$. Evaluate $\lim_{x \rightarrow -1} \frac{xf(x) + 3}{2f(x) + 1}$.

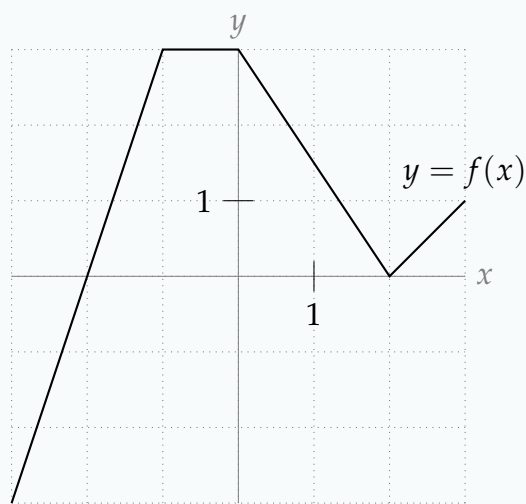
Q[43](*) : Find the value of the constant a for which $\lim_{x \rightarrow -2} \frac{x^2 + ax + 3}{x^2 + x - 2}$ exists.

Q[44]: Suppose $f(x) = 2x$ and $g(x) = \frac{1}{x}$. Evaluate the following limits.

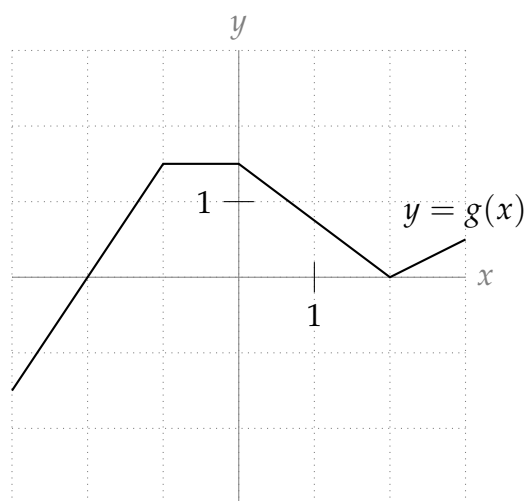
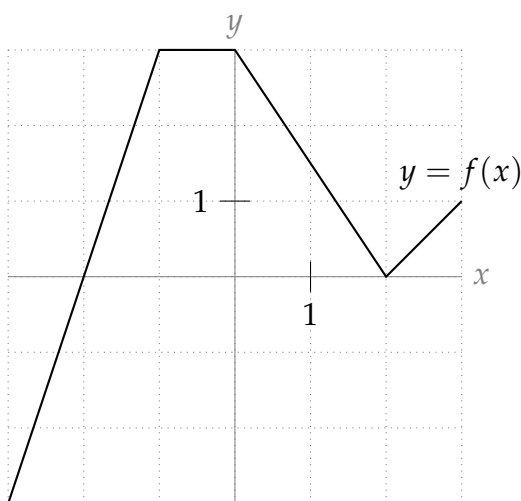
- (a) $\lim_{x \rightarrow 0} f(x)$
- (b) $\lim_{x \rightarrow 0} g(x)$
- (c) $\lim_{x \rightarrow 0} f(x)g(x)$
- (d) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)}$
- (e) $\lim_{x \rightarrow 2} [f(x) + g(x)]$
- (f) $\lim_{x \rightarrow 0} \frac{f(x) + 1}{g(x + 1)}$

► Stage 3

Q[45]: The curve $y = f(x)$ is shown in the graph below. Sketch the graph of $y = \frac{1}{f(x)}$.



Q[46]: The graphs of functions $f(x)$ and $g(x)$ are shown in the graphs below. Use these to sketch the graph of $\frac{f(x)}{g(x)}$.



Q[47]: Suppose the position of a white ball, at time t , is given by $s(t)$, and the position of a red ball is given by $2s(t)$. Using the definition from Section 1.2 of the velocity of a particle, and the limit laws from this section, answer the following question: if the white ball has velocity 5 at time $t = 1$, what is the velocity of the red ball?

Q[48]: Let $f(x) = \frac{1}{x}$ and $g(x) = \frac{-1}{x}$.

(a) Evaluate $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 0} g(x)$.

(b) Evaluate $\lim_{x \rightarrow 0} [f(x) + g(x)]$

(c) Is it always true that $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$?

Q[49]: Suppose

$$f(x) = \begin{cases} x^2 + 3 & , \quad x > 0 \\ 0 & , \quad x = 0 \\ x^2 - 3 & , \quad x < 0 \end{cases}$$

(a) Evaluate $\lim_{x \rightarrow 0^-} f(x)$.

(b) Evaluate $\lim_{x \rightarrow 0^+} f(x)$.

(c) Evaluate $\lim_{x \rightarrow 0} f(x)$.

Q[50]: Suppose

$$f(x) = \begin{cases} \frac{x^2 + 8x + 16}{x^2 + 30x - 4} & , \quad x > -4 \\ x^3 + 8x^2 + 16x & , \quad x \leq -4 \end{cases}$$

(a) Evaluate $\lim_{x \rightarrow -4^-} f(x)$.

(b) Evaluate $\lim_{x \rightarrow -4^+} f(x)$.

(c) Evaluate $\lim_{x \rightarrow -4} f(x)$.

Remark: the average velocity is *not* the average of the two instantaneous velocities.

Solutions to Exercises 1.3 — Jump to [TABLE OF CONTENTS](#)

S-1:

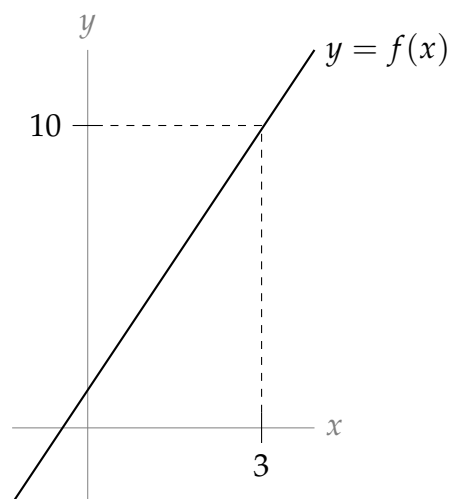
- (a) $\lim_{x \rightarrow -2} f(x) = 1$: as x gets very close to -2 , y gets very close to 1.
- (b) $\lim_{x \rightarrow 0} f(x) = 0$: as x gets very close to 0, y also gets very close to 0.
- (c) $\lim_{x \rightarrow 2} f(x) = 2$: as x gets very close to 2, y gets very close to 2. We ignore the value of the function where x is exactly 2.

S-2: The limit does not exist. As x approaches 0 *from the left*, y approaches -1 ; as x approaches 0 *from the right*, y approaches 1. This tells us $\lim_{x \rightarrow 0^-} f(x) = -1$ and $\lim_{x \rightarrow 0^+} f(x) = 1$, but neither of these are what the question asked. Since the limits from left and right do not agree, the limit does not exist. Put another way, there is no single number y approaches as x approaches 0, so the limit $\lim_{x \rightarrow 0} f(x)$ does not exist.

S-3:

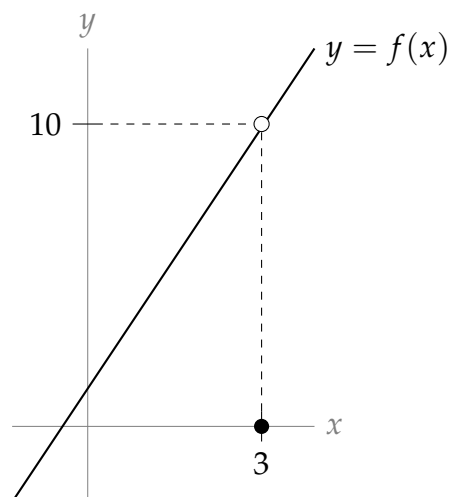
- (a) $\lim_{x \rightarrow -1^-} f(x) = 2$: as x approaches -1 from the left, y approaches 2. It doesn't matter that the function isn't defined at $x = -1$, and it doesn't matter what happens to the right of $x = -1$.
- (b) $\lim_{x \rightarrow -1^+} f(x) = -2$: as x approaches -1 from the right, y approaches -2 . It doesn't matter that the function isn't defined at -1 , and it doesn't matter what happens to the left of -1 .
- (c) $\lim_{x \rightarrow -1} f(x) = \text{DNE}$: since the limits from the left and right don't agree, the limit does not exist.
- (d) $\lim_{x \rightarrow -2^+} f(x) = 0$: as x approaches -2 from the right, y approaches 0. It doesn't matter that the function isn't defined at 2, or to the left of 2.
- (e) $\lim_{x \rightarrow 2^-} f(x) = 0$: as x approaches 2 from the left, y approaches 0. It doesn't matter that the function isn't defined at 2, or to the right of 2.

S-4: Many answers are possible; here is one.



As x gets closer and closer to 3, y gets closer and closer to 10: this shows $\lim_{x \rightarrow 3} f(x) = 10$. Also, at 3 itself, the function takes the value 10; this shows $f(3) = 10$.

S-5: Many answers are possible; here is one.



Note that, as x gets closer and closer to 3 *except at 3 itself*, y gets closer and closer to 10: this shows $\lim_{x \rightarrow 3} f(x) = 10$. Then, when $x = 3$, the function has value 0: this shows $f(3) = 0$.

S-6: In general, this is false. The limit as x goes to 3 does not take into account the value of the function at 3: $f(3)$ can be anything.

S-7: False. The limit as x goes to 3 does not take into account the value of the function at 3: $f(3)$ tells us nothing about $\lim_{x \rightarrow 3} f(x)$.

S-8: $\lim_{x \rightarrow -2^-} f(x) = 16$: in order for the limit $\lim_{x \rightarrow 2} f(x)$ to exist and be equal to 16, both one

sided limits must exist and be equal to 16.

S-9: Not enough information to say. If $\lim_{x \rightarrow -2^+} f(x) = 16$, then $\lim_{x \rightarrow -2} f(x) = 16$. If $\lim_{x \rightarrow -2^+} f(x) \neq 16$, then $\lim_{x \rightarrow -2} f(x)$ does not exist.

S-10: $\lim_{t \rightarrow 0} \sin t = 0$: as t approaches 0, $\sin t$ approaches 0 as well.

S-11: $\lim_{x \rightarrow 0^+} \log x = -\infty$: as x approaches 0 from the right, $\log x$ is negative and increasingly large, growing without bound.

S-12: $\lim_{y \rightarrow 3} y^2 = 9$: as y gets closer and closer to 3, y^2 gets closer and closer to 3^2 .

S-13: $\lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$: as x gets closer and closer to 0 from the left, $\frac{1}{x}$ becomes a larger and larger negative number.

S-14: $\lim_{x \rightarrow 0} \frac{1}{x} = \text{DNE}$: as x gets closer and closer to 0 from the left, $\frac{1}{x}$ becomes a larger and larger negative number; but as x gets closer and closer to 0 from the right, $\frac{1}{x}$ becomes a larger and larger positive number. So the limit from the left is not the same as the limit from the right, and so $\lim_{x \rightarrow 0} \frac{1}{x} = \text{DNE}$. Contrast this with Question **15**.

S-15: $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$: as x gets closer and closer to 0 from the either side, $\frac{1}{x^2}$ becomes a larger and larger positive number, growing without bound. Contrast this with Question **14**.

S-16: $\lim_{x \rightarrow 3} \frac{1}{10} = \frac{1}{10}$: no matter what x is, $\frac{1}{10}$ is always $\frac{1}{10}$. In particular, as x approaches 3, $\frac{1}{10}$ stays put at $\frac{1}{10}$.

S-17: When x is very close to 3, $f(x)$ looks like the function x^2 . So: $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} x^2 = 9$

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S-1: Zeroes cause a problem when they show up in the denominator, so we can only compute [\(a\)](#) and [\(d\)](#). (Both these limits are zero.) Be careful: there is no such rule as “zero divided by zero is one,” or “zero divided by zero is zero.”

S-2: The statement $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = 10$ tells us that, as x gets very close to 3, $f(x)$ is 10 times as large as $g(x)$. We notice that if $f(x) = 10g(x)$, then $\frac{f(x)}{g(x)} = 10$, so $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = 10$ wherever f and g exist. So it's enough to find a function $g(x)$ that has limit 0 at 3. Such a function is (for example) $g(x) = x - 3$. So, we take $f(x) = 10(x - 3)$ and $g(x) = x - 3$. It is easy now to check that $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} g(x) = 0$ and $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 3} \frac{10(x - 3)}{x - 3} = \lim_{x \rightarrow 3} 10 = 10$.

S-3:

- As we saw in Question **2**, $x - 3$ is a function with limit 0 at $x = 3$. So one way of thinking about this question is to try choosing $f(x)$ so that $\frac{f(x)}{g(x)} = g(x) = x - 3$ too, which leads us to the solution $f(x) = (x - 3)^2$ and $g(x) = x - 3$. This is one of many, many possible answers.
- Another way of thinking about this problem is that $f(x)$ should go to 0 “more strongly” than $g(x)$ when x approaches 3. One way of a function going to 0 really strongly is to make that function identically zero. So we can set $f(x) = 0$ and $g(x) = x - 3$. Now $\frac{f(x)}{g(x)}$ is equal to 0 whenever $x \neq 3$, and is undefined at $x = 3$. Since the limit as x goes to three does not take into account the value of the function at 3, we have $\lim_{x \rightarrow 3} \frac{f(x)}{g(x)} = 0$.

There are many more possible answers.

S-4: One way to start this problem is to remember $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$. (Using $\frac{1}{x^2}$ as opposed to $\frac{1}{x}$ is important, since $\lim_{x \rightarrow 0} \frac{1}{x}$ does not exist.) Then by “shifting” by three, we find $\lim_{x \rightarrow 3} \frac{1}{(x - 3)^2} = \infty$. So it is enough to arrange that $\frac{f(x)}{g(x)} = \frac{1}{(x - 3)^2}$. We can achieve this with $f(x) = x - 3$ and $g(x) = (x - 3)^3$, and maintain $\lim_{x \rightarrow 3} f(x) = \lim_{x \rightarrow 3} g(x) = 0$. Again, this is one of many possible solutions.

S-5: Any real number; positive infinity; negative infinity; does not exist.

This is an important thing to remember: often, people see limits that look like $\frac{0}{0}$ and think that the limit must be 1, or 0, or infinite. In fact, this limit could be anything—it depends on the relationship between f and g .

Questions **2** and **3** show us examples where the limit is 10 and 0; they can easily be modified to make the limit any real number.

Question **4** show us an example where the limit is ∞ ; it can easily be modified to make the limit $-\infty$ or DNE.

S-6: Since we're not trying to divide by 0, or multiply by

infinity: $\lim_{t \rightarrow 10} \frac{2(t-10)^2}{t} = \frac{2 \cdot 0}{10} = 0$

S-7: Since we're not doing anything dodgy like putting 0 in the denominator,

$$\lim_{y \rightarrow 0} \frac{(y+1)(y+2)(y+3)}{\cos y} = \frac{(0+1)(0+2)(0+3)}{\cos 0} = \frac{6}{1} = 6.$$

S-8: Since the limits of the numerator and denominator exist, and since the limit of the

denominator is nonzero: $\lim_{x \rightarrow 3} \left(\frac{4x-2}{x+2} \right)^4 = \left(\frac{4(3)-2}{3+2} \right)^4 = 16$

S-9:

$$\lim_{t \rightarrow -3} \left(\frac{1-t}{\cos(t)} \right) = \frac{\lim_{t \rightarrow -3} (1-t)}{\lim_{t \rightarrow -3} \cos(t)} = 4 / \cos(-3) = 4 / \cos(3)$$

S-10: If try naively then we get 0/0, so we expand and then simplify:

$$\frac{(2+h)^2 - 4}{2h} = \frac{h^2 + 4h + 4 - 4}{2h} = \frac{h}{2} + 2$$

Hence the limit is $\lim_{h \rightarrow 0} \left(\frac{h}{2} + 2 \right) = 2.$

S-11:

$$\lim_{t \rightarrow -2} \left(\frac{t-5}{t+4} \right) = \frac{\lim_{t \rightarrow -2} (t-5)}{\lim_{t \rightarrow -2} (t+4)} = -7/2.$$

S-12:

$$\lim_{t \rightarrow 1} \sqrt{5x^3 + 4} = \sqrt{\lim_{t \rightarrow 1} (5x^3 + 4)} = \sqrt{5 \lim_{t \rightarrow 1} (x^3) + 4} = \sqrt{9} = 3.$$

S-13:

$$\lim_{t \rightarrow -1} \left(\frac{t-2}{t+3} \right) = \frac{\lim_{t \rightarrow -1} (t-2)}{\lim_{t \rightarrow -1} (t+3)} = -3/2.$$

S-14: We simply plug in $x = 1$: $\lim_{x \rightarrow 1} \left[\frac{\log(1+x) - x}{x^2} \right] = \log(2) - 1.$

S-15: If we try naively then we get $0/0$, so we simplify first:

$$\frac{x-2}{x^2-4} = \frac{x-2}{(x-2)(x+2)} = \frac{1}{x+2}$$

Hence the limit is $\lim_{x \rightarrow 2} \frac{1}{x+2} = 1/4$.

S-16: If we try to plug in $x = 4$, we find the denominator is zero. So to get a better idea of what's happening, we factor the numerator and denominator:

$$\begin{aligned} \lim_{x \rightarrow 4} \frac{x^2 - 4x}{x^2 - 16} &= \lim_{x \rightarrow 4} \frac{x(x-4)}{(x+4)(x-4)} \\ &= \lim_{x \rightarrow 4} \frac{x}{x+4} \\ &= \frac{4}{8} = \frac{1}{2} \end{aligned}$$

S-17: If we try to plug in $x = 2$, we find the denominator is zero. So to get a better idea of what's happening, we factor the numerator:

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^2 + x - 6}{x - 2} &= \lim_{x \rightarrow 2} \frac{(x+3)(x-2)}{x-2} \\ &= \lim_{x \rightarrow 2} (x+3) = 5 \end{aligned}$$

S-18: If we try naively then we get $0/0$, so we simplify first:

$$\frac{x^2 - 9}{x + 3} = \frac{(x-3)(x+3)}{(x+3)} = x - 3$$

Hence the limit is $\lim_{x \rightarrow -3} (x - 3) = -6$.

S-19: To calculate the limit of a polynomial, we simply evaluate the polynomial:

$$\lim_{t \rightarrow 2} \frac{1}{2}t^4 - 3t^3 + t = \frac{1}{2} \cdot 2^4 - 3 \cdot 2^3 + 2 = -14$$

S-20:

$$\begin{aligned}\frac{\sqrt{x^2+8}-3}{x+1} &= \frac{\sqrt{x^2+8}-3}{x+1} \cdot \frac{\sqrt{x^2+8}+3}{\sqrt{x^2+8}+3} \\&= \frac{(x^2+8)-3^2}{(x+1)(\sqrt{x^2+8}+3)} \\&= \frac{x^2-1}{(x+1)(\sqrt{x^2+8}+3)} \\&= \frac{(x-1)(x+1)}{(x+1)(\sqrt{x^2+8}+3)} \\&= \frac{(x-1)}{\sqrt{x^2+8}+3} \\ \lim_{x \rightarrow -1} \frac{\sqrt{x^2+8}-3}{x+1} &= \lim_{x \rightarrow -1} \frac{(x-1)}{\sqrt{x^2+8}+3} \\&= \frac{-2}{\sqrt{9}+3} \\&= -\frac{2}{6} = -\frac{1}{3}.\end{aligned}$$

S-21: If we try to do the limit naively we get 0/0. Hence we must simplify.

$$\begin{aligned}\frac{\sqrt{x+7}-\sqrt{11-x}}{2x-4} &= \frac{\sqrt{x+7}-\sqrt{11-x}}{2x-4} \cdot \left(\frac{\sqrt{x+7}+\sqrt{11-x}}{\sqrt{x+7}+\sqrt{11-x}} \right) \\&= \frac{(x+7)-(11-x)}{(2x-4)(\sqrt{x+7}+\sqrt{11-x})} \\&= \frac{2x-4}{(2x-4)(\sqrt{x+7}+\sqrt{11-x})} \\&= \frac{1}{\sqrt{x+7}+\sqrt{11-x}} \\ \text{So, } \lim_{x \rightarrow 2} \frac{\sqrt{x+7}-\sqrt{11-x}}{2x-4} &= \lim_{x \rightarrow 2} \frac{1}{\sqrt{x+7}+\sqrt{11-x}} \\&= \frac{1}{\sqrt{9}+\sqrt{9}} \\&= \frac{1}{6}\end{aligned}$$

S-22: If we try to do the limit naively we get $0/0$. Hence we must simplify.

$$\begin{aligned}\frac{\sqrt{x+2}-\sqrt{4-x}}{x-1} &= \frac{\sqrt{x+2}-\sqrt{4-x}}{x-1} \cdot \frac{\sqrt{x+2}+\sqrt{4-x}}{\sqrt{x+2}+\sqrt{4-x}} \\ &= \frac{(x+2)-(4-x)}{(x-1)(\sqrt{x+2}+\sqrt{4-x})} \\ &= \frac{2x-2}{(x-1)(\sqrt{x+2}+\sqrt{4-x})} \\ &= \frac{2}{\sqrt{x+2}+\sqrt{4-x}}\end{aligned}$$

So the limit is

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{\sqrt{x+2}-\sqrt{4-x}}{x-1} &= \lim_{x \rightarrow 1} \frac{2}{\sqrt{x+2}+\sqrt{4-x}} \\ &= \frac{2}{\sqrt{3}+\sqrt{3}} \\ &= \frac{1}{\sqrt{3}}\end{aligned}$$

S-23: If we try to do the limit naively we get $0/0$. Hence we must simplify.

$$\begin{aligned}\frac{\sqrt{x-2}-\sqrt{4-x}}{x-3} &= \frac{\sqrt{x-2}-\sqrt{4-x}}{x-3} \cdot \frac{\sqrt{x-2}+\sqrt{4-x}}{\sqrt{x-2}+\sqrt{4-x}} \\ &= \frac{(x-2)-(4-x)}{(x-3)(\sqrt{x-2}+\sqrt{4-x})} \\ &= \frac{2x-6}{(x-3)(\sqrt{x-2}+\sqrt{4-x})} \\ &= \frac{2}{\sqrt{x-2}+\sqrt{4-x}} \\ \text{So, } \lim_{x \rightarrow 3} \frac{\sqrt{x-2}-\sqrt{4-x}}{x-3} &= \lim_{x \rightarrow 3} \frac{2}{\sqrt{x-2}+\sqrt{4-x}} \\ &= \frac{2}{1+1} \\ &= 1.\end{aligned}$$

S-24: Here we get 0/0 if we try the naive approach. Hence we must simplify.

$$\begin{aligned}\frac{3t-3}{2-\sqrt{5-t}} &= \frac{3t-3}{2-\sqrt{5-t}} \times \frac{2+\sqrt{5-t}}{2+\sqrt{5-t}} \\ &= (2+\sqrt{5-t}) \frac{3t-3}{2^2-(5-t)} \\ &= (2+\sqrt{5-t}) \frac{3t-3}{t-1} \\ &= (2+\sqrt{5-t}) \frac{3(t-1)}{t-1}\end{aligned}$$

So there is a cancelation. Hence the limit is

$$\begin{aligned}\lim_{t \rightarrow 1} \frac{3t-3}{2-\sqrt{5-t}} &= \lim_{t \rightarrow 1} (2+\sqrt{5-t}) \cdot 3 \\ &= 12\end{aligned}$$

S-25: This is a classic example of the Squeeze Theorem. It is tempting to try to use arithmetic of limits: $\lim_{x \rightarrow 0} -x^2 = 0$, and $\lim_{x \rightarrow 0} \cos\left(\frac{3}{x}\right) = \text{something}$, and zero times something is 0. However, this is invalid reasoning, because we can only use arithmetic of limits when those limits exist, and $\lim_{x \rightarrow 0} \cos\left(\frac{3}{x}\right)$ does not exist. So, we need the Squeeze Theorem.

Since $-1 \leq \cos\left(\frac{3}{x}\right) \leq 1$, we can bound our function of interest from above and below (being careful of the sign!):

$$-x^2(\mathbf{1}) \leq -x^2 \cos\left(\frac{3}{x}\right) \leq -x^2(\mathbf{-1})$$

So our function of interest is between $-x^2$ and x^2 . Since $\lim_{x \rightarrow 0} -x^2 = \lim_{x \rightarrow 0} x^2 = 0$, by the

Squeeze Theorem, also $\lim_{x \rightarrow 0} -x^2 \cos\left(\frac{3}{x}\right) = 0$.

Advice about writing these up: whenever we use the Squeeze Theorem, we need to explicitly write that two things are true: that the function we're interested is bounded above and below by two other functions, and that both of those functions have the same limit. Then we can conclude (and we need to write this down as well!) that our original function also shares that limit.

S-26: Recall that sine and cosine, no matter what (real-number) input we feed them, spit out numbers between -1 and 1 . So we can bound our horrible numerator, rather than trying to deal with it directly.

$$x^4(\mathbf{-1}) + 5x^2(\mathbf{-1}) + 2 \leq x^4 \sin\left(\frac{1}{x}\right) + 5x^2 \cos\left(\frac{1}{x}\right) + 2 \leq x^4(\mathbf{1}) + 5x^2(\mathbf{1}) + 2$$

Further, notice that our bounded functions tend to the same value as x goes to 0:

$$\lim_{x \rightarrow 0} x^4(-1) + 5x^2(-1) + 2 = \lim_{x \rightarrow 0} x^4 + 5x^2 + 2 = 2.$$

So, by the Squeeze Theorem, also

$$\lim_{x \rightarrow 0} x^4 \sin\left(\frac{1}{x}\right) + 5x^2 \cos\left(\frac{1}{x}\right) + 2 = 2.$$

Now, we evaluate our original limit:

$$\lim_{x \rightarrow 0} \frac{x^4 \sin\left(\frac{1}{x}\right) + 5x^2 \cos\left(\frac{1}{x}\right) + 2}{(x-2)^2} = \frac{2}{(-2)^2} = \frac{1}{2}.$$

S-27: $\lim_{x \rightarrow 0} \sin^2\left(\frac{1}{x}\right) = DNE$, so we think about using the Squeeze Theorem. We'll need to bound the expression $x \sin^2\left(\frac{1}{x}\right)$, but the bounding is a little delicate. For any non-zero value we plug in for x , $\sin^2\left(\frac{1}{x}\right)$ is a number in the interval $[0, 1]$. If a is a number in the interval $[0, 1]$, then:

$$\begin{array}{ll} 0 \leq xa \leq x & \text{when } x \text{ is positive} \\ x \leq xa \leq 0 & \text{when } x \text{ is negative} \end{array}$$

We'll show you two ways to use this information to create bounds that will allow you to apply the Squeeze Theorem.

- Solution 1: We will evaluate separately the limit from the right and from the left.

When $x > 0$,

$$0 \leq x \sin^2\left(\frac{1}{x}\right) \leq x$$

because $0 \leq \sin^2\left(\frac{1}{x}\right) \leq 1$. Since

$$\lim_{x \rightarrow 0^+} 0 = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^+} x = 0$$

then by the Squeeze Theorem, also

$$\lim_{x \rightarrow 0^+} x \sin^2\left(\frac{1}{x}\right) = 0.$$

Similarly, When $x < 0$,

$$x \leq x \sin^2\left(\frac{1}{x}\right) \leq 0$$

because $0 \leq \sin^2\left(\frac{1}{x}\right) \leq 1$. Since

$$\lim_{x \rightarrow 0^-} x = 0 \quad \text{and} \quad \lim_{x \rightarrow 0^-} 0 = 0$$

then by the Squeeze Theorem, also

$$\lim_{x \rightarrow 0^-} x \sin^2\left(\frac{1}{x}\right) = 0.$$

Since the one-sided limits are both equal to zero,

$$\lim_{x \rightarrow 0} x \sin^2\left(\frac{1}{x}\right) = 0.$$

Remark: this is a perfectly fine proof, but it seems to repeat itself. Since the cases $x < 0$ and $x > 0$ are so similar, we would like to take care of them together. This can be done as shown below.

- Solution 2: If $x \neq 0$, then $0 \leq \sin^2\left(\frac{1}{x}\right) \leq 1$, so

$$-|x| \leq x \sin^2\left(\frac{1}{x}\right) \leq |x|.$$

Since

$$\lim_{x \rightarrow 0} -|x| = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} |x| = 0$$

then by the Squeeze Theorem, also

$$\lim_{x \rightarrow 0} x \sin^2\left(\frac{1}{x}\right) = 0.$$

S-28: When we plug $w = 5$ in to the numerator and denominator, we find that each becomes zero. Since we can't divide by zero, we have to dig a little deeper. When a polynomial has a root, that also means it has a factor: we can factor $(w - 5)$ out of the top. That lets us cancel:

$$\lim_{w \rightarrow 5} \frac{2w^2 - 50}{(w - 5)(w - 1)} = \lim_{w \rightarrow 5} \frac{2(w - 5)(w + 5)}{(w - 5)(w - 1)} = \lim_{w \rightarrow 5} \frac{2(w + 5)}{(w - 1)}.$$

Note that the function $\frac{2w^2 - 50}{(w - 5)(w - 1)}$ is NOT defined at $w = 5$, while the function $\frac{2(w + 5)}{(w - 1)}$ IS defined at $w = 5$; so strictly speaking, these two functions are not equal. However, for every value of w that is not 5, the functions are the same, so their *limits* are

equal. Furthermore, the limit of the second function is quite easy to calculate, since we've eliminated the zero in the denominator: $\lim_{w \rightarrow 5} \frac{2(w+5)}{(w-1)} = \frac{2(5+5)}{5-1} = 5$.

So $\lim_{w \rightarrow 5} \frac{2w^2 - 50}{(w-5)(w-1)} = \lim_{w \rightarrow 5} \frac{2(w+5)}{(w-1)} = 5$.

S-29: When we plug in $r = -5$ to the denominator, we find that it becomes 0, so we need to dig deeper. The numerator is not zero, so cancelling is out. Notice that the denominator is factorable: $r^2 + 10r + 25 = (r+5)^2$. As r approaches -5 from either side, the denominator gets very close to zero, but stays positive. The numerator gets very close to -5 . So, as r gets closer to -5 , we have something close to -5 divided by a very small, positive number. Since the denominator is small, the fraction will have a large magnitude; since the numerator is negative and the denominator is positive, the fraction will be negative. So, $\lim_{r \rightarrow -5} \frac{r}{r^2 + 10r + 25} = -\infty$

S-30: First, we find $\lim_{x \rightarrow -1} \frac{x^3 + x^2 + x + 1}{3x + 3}$. When we plug in $x = -1$ to the top and the bottom, both become zero. In a polynomial, where there is a root, there is a factor, so this tells us we can factor out $(x+1)$ from both the top and the bottom. It's pretty easy to see how to do this in the bottom. For the top, if you're having a hard time, one factoring method (of many) to try is long division of polynomials; another is to factor out $(x+1)$ from the first two terms and the last two terms. (Detailed examples of long division are given in Appendix A.16 and Examples 1.10.2 and 1.10.3 of the CLP-2 text.)

$$\begin{aligned} \lim_{x \rightarrow -1} \frac{x^3 + x^2 + x + 1}{3x + 3} &= \lim_{x \rightarrow -1} \frac{x^2(x+1) + (x+1)}{3x+3} = \lim_{x \rightarrow -1} \frac{(x+1)(x^2+1)}{3(x+1)} \\ &= \lim_{x \rightarrow -1} \frac{x^2+1}{3} = \frac{(-1)^2+1}{3} = \frac{2}{3}. \end{aligned}$$

One thing to note here is that the function $\frac{x^3 + x^2 + x + 1}{3x + 3}$ is not defined at $x = -1$

(because we can't divide by zero). So we replaced it with the function $\frac{x^2+1}{3}$, which IS defined at $x = -1$. These functions only differ at $x = -1$; they are the same at every other point. That is why we can use the second function to find the limit of the first function.

Now we're ready to find the actual limit asked in the problem:

$$\lim_{x \rightarrow -1} \sqrt{\frac{x^3 + x^2 + x + 1}{3x + 3}} = \sqrt{\frac{2}{3}}.$$

S-31: When we plug $x = 0$ into the denominator, we get 0, which means we need to look harder. The numerator is not zero, so we won't be able to cancel our problems away. Let's factor to make things clearer.

$$\frac{x^2 + 2x + 1}{3x^5 - 5x^3} = \frac{(x + 1)^2}{x^3(3x^2 - 5)}$$

As x gets close to 0, the numerator is close to 1; the term $(3x^2 - 5)$ is negative; and the sign of x^3 depends on the direction we're approaching 0 from. Since we're dividing a numerator that is very close to 1 by something that's getting very close to 0, the magnitude of the fraction is getting bigger and bigger without bound. Since the sign of the fraction flips depending on whether we are using numbers slightly bigger than 0, or slightly smaller than 0, that means the one-sided limits are ∞ and $-\infty$, respectively. (In particular, $\lim_{x \rightarrow 0^-} \frac{x^2 + 2x + 1}{3x^5 - 5x^3} = \infty$ and $\lim_{x \rightarrow 0^+} \frac{x^2 + 2x + 1}{3x^5 - 5x^3} = -\infty$.) Since the one-sided limits don't agree, the limit does not exist.

S-32: As usual, we first try plugging in $t = 7$, but the denominator is 0, so we need to think harder. The top and bottom are both squares, so let's go ahead and factor:

$\frac{t^2x^2 + 2tx + 1}{t^2 - 14t + 49} = \frac{(tx + 1)^2}{(t - 7)^2}$. Since x is positive, the numerator is nonzero. Also, the numerator is positive near $t = 7$. So, we have something positive and nonzero on the top, and we divide it by the bottom, which is positive and getting closer and closer to zero. The quotient is always positive near $t = 7$, and it is growing in magnitude without bound, so $\lim_{t \rightarrow 7} \frac{t^2x^2 + 2tx + 1}{t^2 - 14t + 49} = \infty$.

Remark: there is an important reason we specified that x must be a *positive* constant. Suppose x were $-\frac{1}{7}$ (which is negative and so was not allowed in the question posed). In this case, we would have

$$\begin{aligned} \lim_{t \rightarrow 7} \frac{t^2x^2 + 2tx + 1}{t^2 - 14t + 49} &= \lim_{t \rightarrow 7} \frac{(tx + 1)^2}{(t - 7)^2} \\ &= \lim_{t \rightarrow 7} \frac{(-t/7 + 1)^2}{(t - 7)^2} \\ &= \lim_{t \rightarrow 7} \frac{(-1/7)^2(t - 7)^2}{(t - 7)^2} \\ &= \lim_{t \rightarrow 7} (-1/7)^2 \\ &= \frac{1}{49} \\ &\neq \infty \end{aligned}$$

S-33: The function whose limit we are taking does not depend on d . Since x is a constant,

$x^5 - 32x + 15$ is also a constant—it's just some number, that doesn't change, regardless of what d does. So $\lim_{d \rightarrow 0} x^5 - 32x + 15 = x^5 - 32x + 15$.

S-34: There's a lot going on inside that sine function... and we don't have to care about any of it. No matter what horrible thing we put inside a sine function, the sine function will spit out a number between -1 and 1 . So that means we can bound our horrible function like this:

$$(x-1)^2 \cdot (-1) \leq (x-1)^2 \sin \left[\left(\frac{x^2 - 3x + 2}{x^2 - 2x + 1} \right)^2 + 15 \right] \leq (x-1)^2 \cdot (1)$$

Since $\lim_{x \rightarrow 1} (x-1)^2 \cdot (-1) = \lim_{x \rightarrow 1} (x-1)^2 \cdot (1) = 0$, the Squeeze Theorem tells us that

$$\lim_{x \rightarrow 1} (x-1)^2 \sin \left[\left(\frac{x^2 - 3x + 2}{x^2 - 2x + 1} \right)^2 + 15 \right] = 0$$

as well.

S-35: Since $-1 \leq \sin x \leq 1$ for all values of x ,

$$\begin{aligned} -1 &\leq \sin(x^{-100}) \leq 1 \\ (-1)x^{1/101} &\leq x^{1/101} \sin(x^{-100}) \leq (1)x^{1/101} \quad \text{when } x > 0, \text{ and} \\ (1)x^{1/101} &\leq x^{1/101} \sin(x^{-100}) \leq (-1)x^{1/101} \quad \text{when } x < 0. \text{ Also,} \\ \lim_{x \rightarrow 0} x^{1/101} &= \lim_{x \rightarrow 0} -x^{1/101} = 0 \quad \text{So, by the Squeeze Theorem,} \\ \lim_{x \rightarrow 0^-} x^{1/101} \sin(x^{-100}) &= 0 = \lim_{x \rightarrow 0^+} x^{1/101} \sin(x^{-100}) \quad \text{and so} \\ 0 &= \lim_{x \rightarrow 0} x^{1/101} \sin(x^{-100}). \end{aligned}$$

Remark: there is a technical point here. When x is a positive number, $-x$ is negative, so $(-1)x < x$. But, when x is negative, $-x$ is positive, so $(-1)x > x$. This is why we take the one-sided limits of our function, and apply the Squeeze Theorem to them separately. It is *not true* to say that, for instance, $(-1)x^{1/101} \leq x^{1/101} \sin(x^{-100})$ when x is near zero, because this does not hold when x is less than zero.

S-36:

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x^2 - 2x} = \lim_{x \rightarrow 2} \frac{(x-2)(x+2)}{x(x-2)} = \lim_{x \rightarrow 2} \frac{x+2}{x} = 2$$

S-37: When we plug in $x = 5$ to the top and the bottom, both limits exist, and the bottom is nonzero. So $\lim_{x \rightarrow 5} \frac{(x-5)^2}{x+5} = \frac{0}{10} = 0$.

S-38: Since we can't plug in $t = \frac{1}{2}$, we'll simplify. One way to start is to add the fractions in the numerator. We'll need a common demoninator, such as $3t^2(t^2 - 1)$.

$$\begin{aligned}\lim_{t \rightarrow \frac{1}{2}} \frac{\frac{1}{3t^2} + \frac{1}{t^2-1}}{2t-1} &= \lim_{t \rightarrow \frac{1}{2}} \frac{\frac{t^2-1}{3t^2(t^2-1)} + \frac{3t^2}{3t^2(t^2-1)}}{2t-1} \\ &= \lim_{t \rightarrow \frac{1}{2}} \frac{\frac{4t^2-1}{3t^2(t^2-1)}}{2t-1} \\ &= \lim_{t \rightarrow \frac{1}{2}} \frac{4t^2-1}{3t^2(t^2-1)(2t-1)} \\ &= \lim_{t \rightarrow \frac{1}{2}} \frac{(2t+1)(2t-1)}{3t^2(t^2-1)(2t-1)} \\ &= \lim_{t \rightarrow \frac{1}{2}} \frac{2t+1}{3t^2(t^2-1)}\end{aligned}$$

Since we cancelled out the term that was causing the numerator and denominator to be zero when $t = \frac{1}{2}$, now $t = \frac{1}{2}$ is in the domain of our function, so we simply plug it in:

$$\begin{aligned}&= \frac{1+1}{\frac{3}{4}\left(\frac{1}{4}-1\right)} \\ &= \frac{2}{\frac{3}{4}\left(-\frac{3}{4}\right)} \\ &= -\frac{32}{9}\end{aligned}$$

S-39: We recall that

$$|x| = \begin{cases} x & , \quad x \geq 0 \\ -x & , \quad x < 0 \end{cases}$$

So,

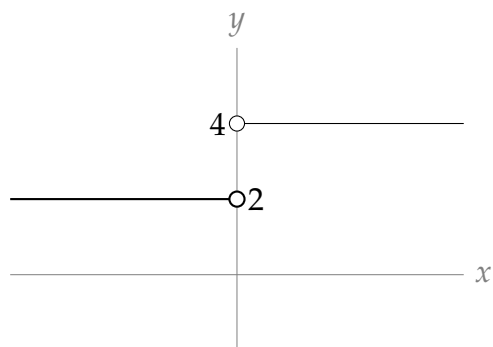
$$\begin{aligned}\frac{|x|}{x} &= \begin{cases} \frac{x}{x} & , \quad x > 0 \\ \frac{-x}{x} & , \quad x < 0 \end{cases} \\ &= \begin{cases} 1 & , \quad x > 0 \\ -1 & , \quad x < 0 \end{cases}\end{aligned}$$

Therefore,

$$3 + \frac{|x|}{x} = \begin{cases} 4 & , \quad x > 0 \\ 2 & , \quad x < 0 \end{cases}$$

Since our function gives a value of 4 when x is to the right of zero, and a value of 2 when x is to the left of zero, $\lim_{x \rightarrow 0} \left(3 + \frac{|x|}{x}\right)$ does not exist.

To further clarify the situation, the graph of $y = f(x)$ is sketched below:



S-40: If we factor out 3 from the numerator, our function becomes $3 \frac{|d+4|}{d+4}$. We recall that

$$|X| = \begin{cases} X & , \quad X \geq 0 \\ -X & , \quad X < 0 \end{cases}$$

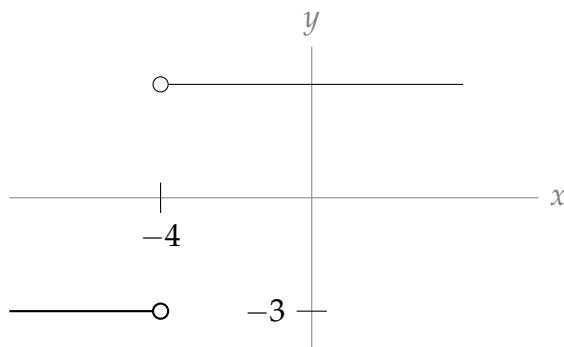
So, with $X = d + 4$,

$$\begin{aligned} 3 \frac{|d+4|}{d+4} &= \begin{cases} 3 \frac{d+4}{d+4} & , \quad d+4 > 0 \\ 3 \frac{-(d+4)}{d+4} & , \quad d+4 < 0 \end{cases} \\ &= \begin{cases} 3 & , \quad d > -4 \\ -3 & , \quad d < -4 \end{cases} \end{aligned}$$

Since our function gives a value of 3 when $d > -4$, and a value of -3 when $d < -4$,

$\lim_{d \rightarrow -4} \frac{|3d+12|}{d+4}$ does not exist.

To further clarify the situation, the graph of $y = f(x)$ is sketched below:



S-41: Note that $x = 0$ is in the domain of our function, and nothing “weird” is happening there: we aren’t dividing by zero, or taking the square root of a negative number, or joining two pieces of a piecewise-defined function. So, as x gets extremely close to zero, $\frac{5x-9}{|x|+2}$ is getting extremely close to $\frac{0-9}{0+2} = -\frac{9}{2}$.

That is, $\lim_{x \rightarrow 0} \frac{5x-9}{|x|+2} = -\frac{9}{2}$.

S-42: Since we aren't dividing by zero, and all these limits exist:

$$\lim_{x \rightarrow -1} \frac{xf(x) + 3}{2f(x) + 1} = \frac{(-1)(-1) + 3}{2(-1) + 1} = -4.$$

S-43: As $x \rightarrow -2$, the denominator goes to 0, and the numerator goes to $-2a + 7$. For the ratio to have a limit, the numerator must also converge to 0, so we need $a = \frac{7}{2}$. Then,

$$\begin{aligned} \lim_{x \rightarrow -2} \frac{x^2 + ax + 3}{x^2 + x - 2} &= \lim_{x \rightarrow -2} \frac{x^2 + \frac{7}{2}x + 3}{(x+2)(x-1)} \\ &= \lim_{x \rightarrow -2} \frac{(x+2)(x + \frac{3}{2})}{(x+2)(x-1)} \\ &= \lim_{x \rightarrow -2} \frac{x + \frac{3}{2}}{x-1} \\ &= \frac{1}{6} \end{aligned}$$

so the limit exists when $a = \frac{7}{2}$.

S-44:

(a) $\lim_{x \rightarrow 0} f(x) = 0$: as x approaches 0, so does $2x$.

(b) $\lim_{x \rightarrow 0} g(x) = \text{DNE}$: the left and right limits do not agree, so the limit does not exist. In particular: $\lim_{x \rightarrow 0^-} g(x) = -\infty$ and $\lim_{x \rightarrow 0^+} g(x) = \infty$.

(c) $\lim_{x \rightarrow 0} f(x)g(x) = \lim_{x \rightarrow 0} 2x \cdot \frac{1}{x} = \lim_{x \rightarrow 0} 2 = 2$.

Remark: although the limit of $g(x)$ does not exist here, the limit of $f(x)g(x)$ does.

(d) $\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{2x}{\frac{1}{x}} = \lim_{x \rightarrow 0} 2x^2 = 0$

(e) $\lim_{x \rightarrow 2} f(x) + g(x) = \lim_{x \rightarrow 2} 2x + \frac{1}{x} = 4 + \frac{1}{2} = \frac{9}{2}$

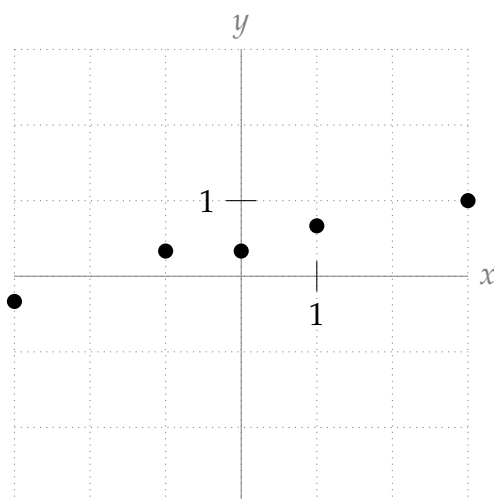
(f) $\lim_{x \rightarrow 0} \frac{f(x) + 1}{g(x + 1)} = \lim_{x \rightarrow 0} \frac{2x + 1}{\frac{1}{x+1}} = \frac{1}{1} = 1$

S-45: We can begin by plotting the points that are easy to read off the diagram.

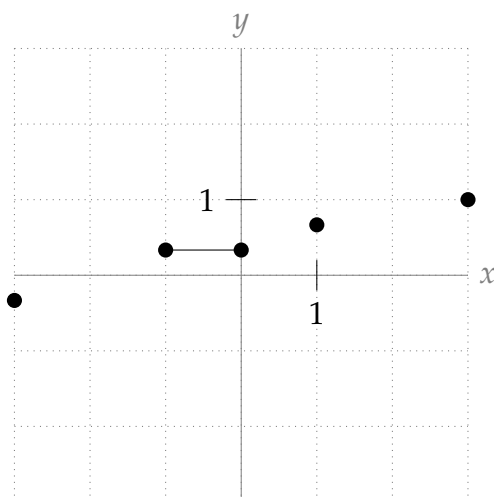
x	$f(x)$	$\frac{1}{f(x)}$
-3	-3	$-\frac{1}{3}$
-2	0	UND
-1	3	$\frac{1}{3}$
0	3	$\frac{1}{3}$
1	$\frac{3}{2}$	$\frac{2}{3}$
2	0	UND
3	1	1

Note that $\frac{1}{f(x)}$ is undefined when $f(x) = 0$. So $\frac{1}{f(x)}$ is undefined at $x = -2$ and $x = 2$. We shall look more closely at the behaviour of $\frac{1}{f(x)}$ for x near ± 2 shortly.

Plotting the above points, we get the following picture:

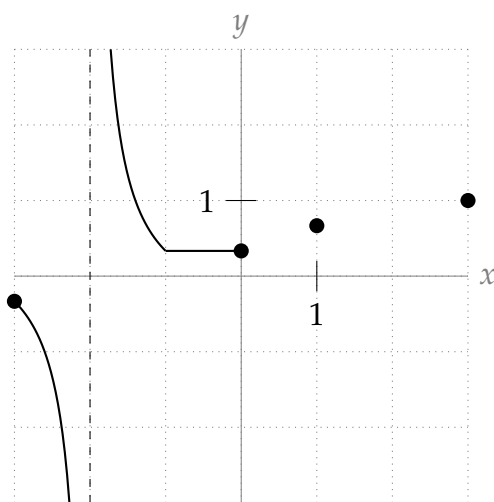


Since $f(x)$ is constant when x is between -1 and 0, then also $\frac{1}{f(x)}$ is constant between -1 and 0, so we update our picture:

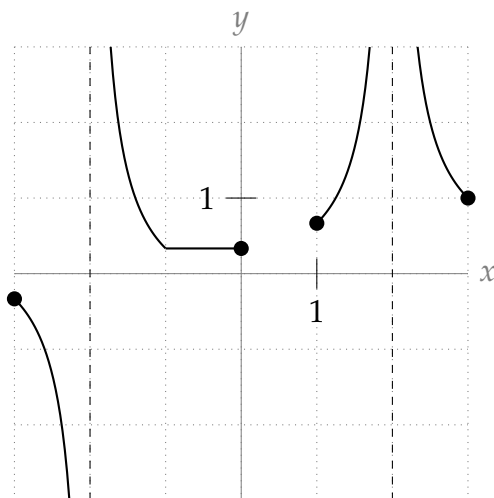


The big question that remains is the behaviour of $\frac{1}{f(x)}$ when x is near -2 and 2. We can

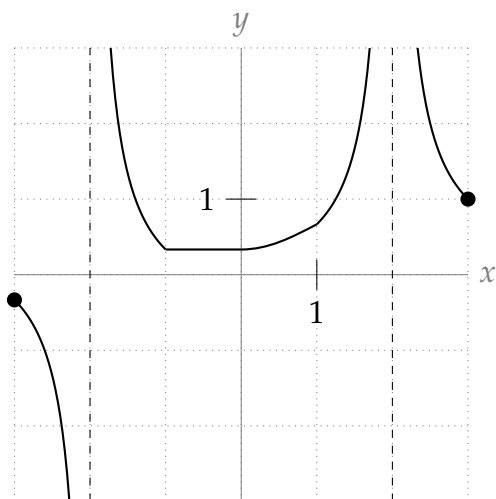
answer this question with limits. As x approaches -2 from the *left*, $f(x)$ gets closer to zero, and is negative. So $\frac{1}{f(x)}$ will be negative, and will increase in magnitude without bound; that is, $\lim_{x \rightarrow -2^-} \frac{1}{f(x)} = -\infty$. Similarly, as x approaches -2 from the *right*, $f(x)$ gets closer to zero, and is positive. So $\frac{1}{f(x)}$ will be positive, and will increase in magnitude without bound; that is, $\lim_{x \rightarrow -2^+} \frac{1}{f(x)} = \infty$. We add this behaviour to our graph:



Now, we consider the behaviour at $x = 2$. Since $f(x)$ gets closer and closer to 0 AND is positive as x approaches 2, we conclude $\lim_{x \rightarrow 2} \frac{1}{f(x)} = \infty$. Adding to our picture:



Now the only remaining blank space is between $x = 0$ and $x = 1$. Since $f(x)$ is a smooth curve that stays away from 0, we can draw some kind of smooth curve here, and call it good enough. (Later on we'll go into more details about drawing graphs. The purpose of this exercise was to utilize what we've learned about limits.)

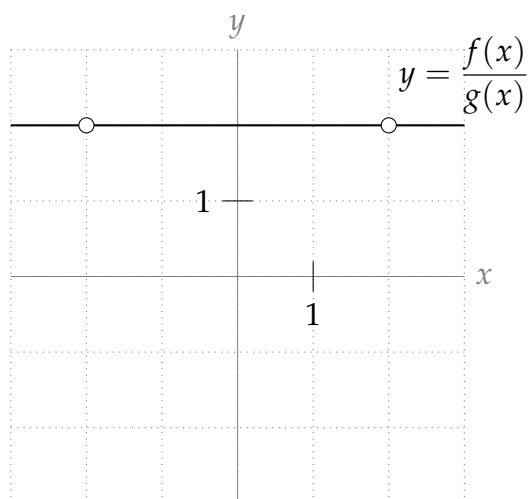


S-46: We can start by examining points.

x	$f(x)$	$g(x)$	$\frac{f(x)}{g(x)}$
-3	-3	-1.5	2
-2	0	0	UND
-1	3	1.5	2
-0	3	1.5	2
1	1.5	.75	2
2	0	0	UND
3	1	.5	2

We cannot divide by zero, so $\frac{f(x)}{g(x)}$ is not defined when $x = \pm 2$. But for every other value of x that we plotted, $f(x)$ is twice as large as $g(x)$, $\frac{f(x)}{g(x)} = 2$. With this in mind, we see that the graph of $f(x)$ is exactly the graph of $2g(x)$.

This gives us the graph below.



Remark: $f(2) = g(2) = 0$, so $\frac{f(2)}{g(2)}$ does not exist, but $\lim_{x \rightarrow 2} \frac{f(x)}{g(x)} = 2$. Although we are trying to “divide by zero” at $x = \pm 2$, it would be a mistake here to interpret this as a vertical asymptote.

S-47: Velocity of white ball when $t = 1$ is $\lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h}$, so the given information tells us $\lim_{h \rightarrow 0} \frac{s(1+h) - s(1)}{h} = 5$. Then the velocity of the red ball when $t = 1$ is $\lim_{h \rightarrow 0} \frac{2s(1+h) - 2s(1)}{h} = \lim_{h \rightarrow 0} 2 \cdot \frac{s(1+h) - s(1)}{h} = 2 \cdot 5 = 10$.

S-48: (a) Neither limit exists. When x gets close to 0, these limits go to positive infinity from one side, and negative infinity from the other.

(b) $\lim_{x \rightarrow 0} [f(x) + g(x)] = \lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{1}{x} \right] = \lim_{x \rightarrow 0} 0 = 0$.

(c) No: this is an example of a time when the two individual functions have limits that don't exist, but the limit of their sum does exist. This “sum rule” is only true when both $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist.

S-49: (a) When we evaluate the limit from the left, we only consider values of x that are less than zero. For these values of x , our function is $x^2 - 3$. So,

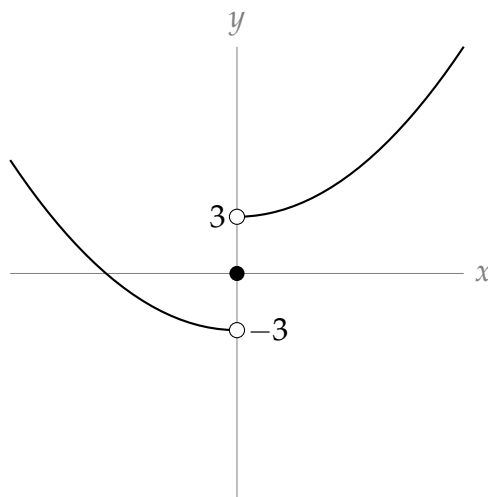
$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^2 - 3) = -3.$$

(b) When we evaluate the limit from the right, we only consider values of x that are greater than zero. For these values of x , our function is $x^2 + 3$. So,

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (x^2 + 3) = 3.$$

(c) Since the limits from the left and right do not agree, $\lim_{x \rightarrow 0} f(x) = \text{DNE}$.

To further clarify the situation, the graph of $y = f(x)$ is sketched below:



S-50: (a) When we evaluate $\lim_{x \rightarrow -4^-} f(x)$, we only consider values of x that are less than